

# Lecture 3: Computational Cameras

## Digital Photography

Instructor: Katie Bouman

TAs: Brandon Zhao, Brady Bhalla, Zachary Huang

# Course announcements

- If you took a Pi but didn't record it with a TA you need to let me know. If you took a Pi but are not officially enrolled let me know.
- Don't have a Pi yet? Pick up at office hours
- Homework due 10 am Thursday on Gradescope

# Slide credits

Many of these slides/images were borrowed/adapted from:

- Ioannis (Yannis) Gkioulekas (CMU)
- Marc Levoy (Stanford)

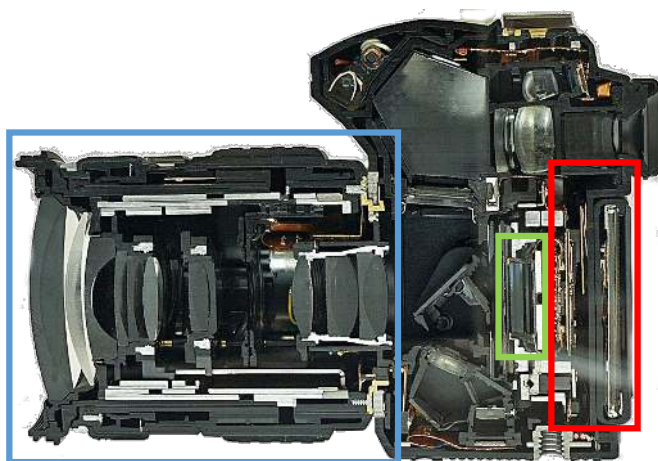
# The modern photography pipeline



# The modern photography pipeline



post-capture processing



optics and  
optical controls

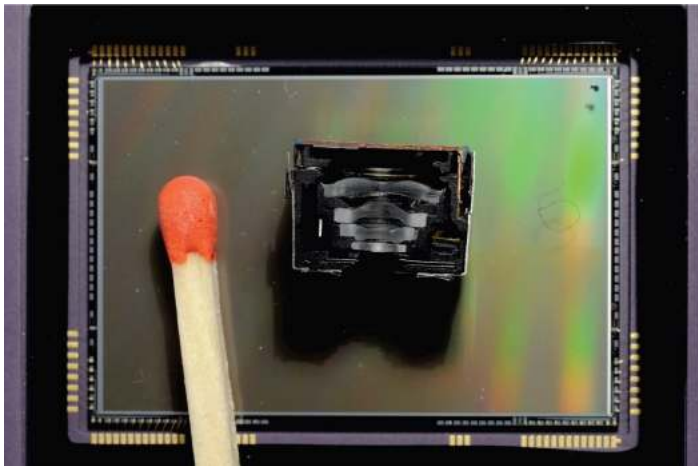
sensor, analog  
front-end, and  
color filter array

in-camera image  
processing  
pipeline

# The modern photography pipeline



post-capture processing



4 element mobile phone lens (on 24x36mm sensor)

optics and  
optical controls



sensor, analog  
front-end, and  
color filter array

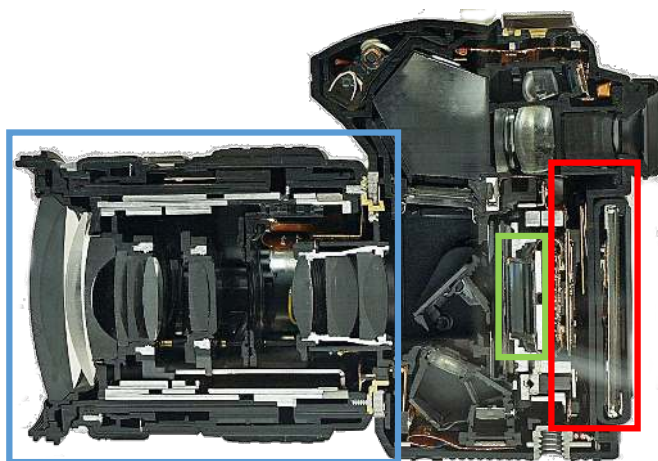


in-camera image  
processing  
pipeline

# The modern photography pipeline



post-capture processing



optics and  
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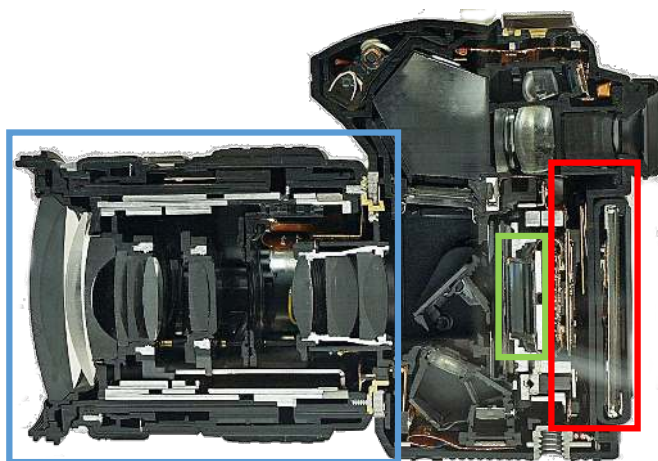
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post-capture processing



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sensor, analog  
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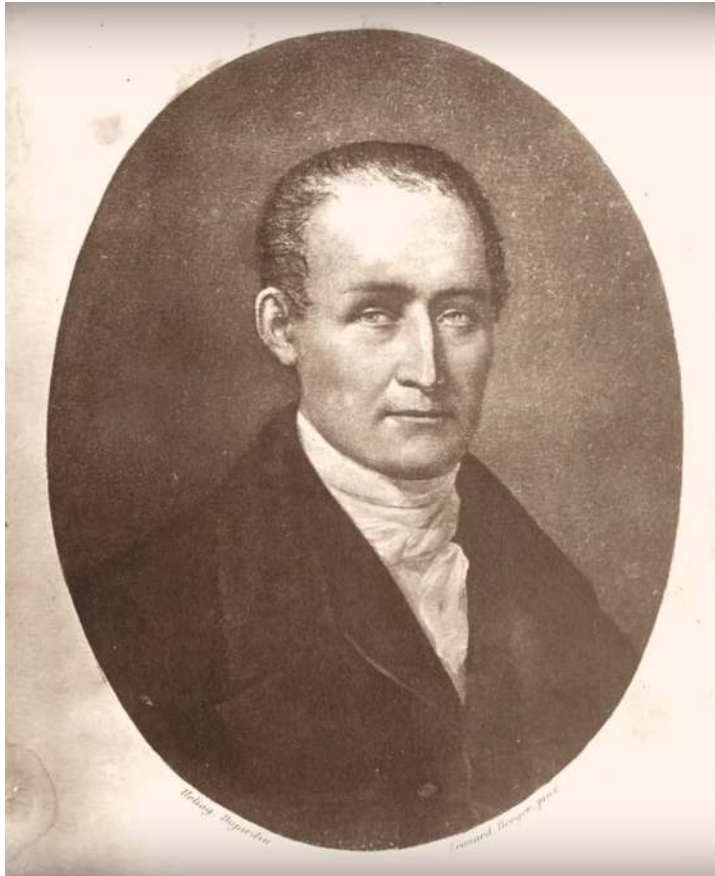


in-camera image  
processing  
pipeline



# Imaging sensor primer

# The oldest photograph



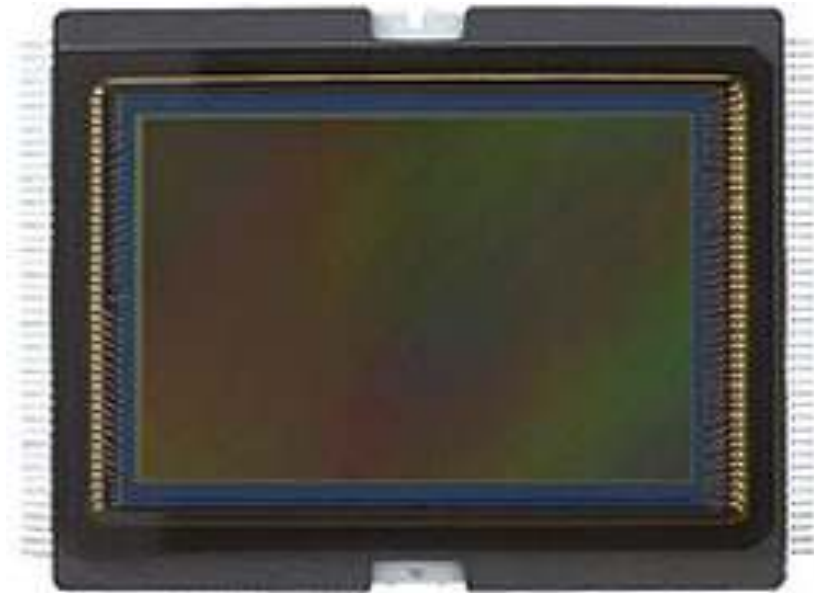
Joseph Nicéphore Niépce



oldest known camera photograph still in existence  
(captured with asphalt on a pewter plate in 1826)

# Imaging sensors

- Two main types of electronic image sensors  
Charge-coupled device (CCD) and complementary metal oxide semiconductor (CMOS)

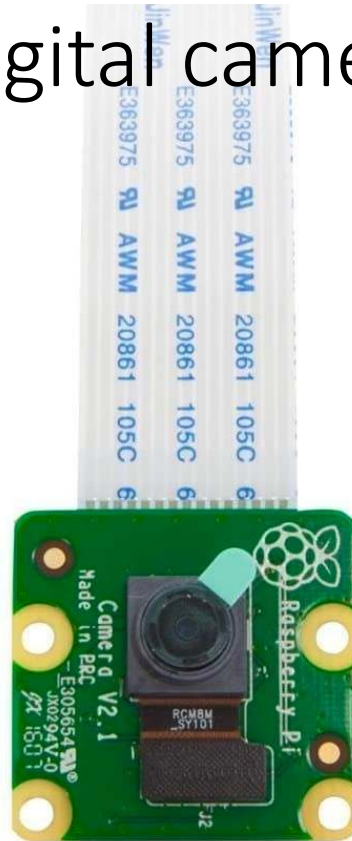


Canon 6D sensor  
(20.20 MP, full-frame)

# Modern digital cameras



Canon 6D Mark II - \$1,400  
+ (\$900 lens)



Raspberry Pi Camera - \$29



Phone



# First digital cameras

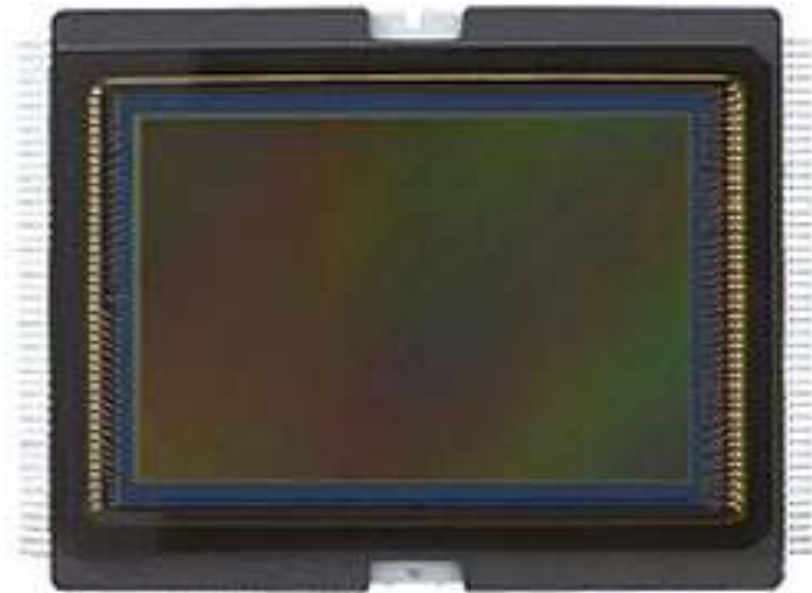


First electronic camera that captured on a CCD  
(but wrote out to analog)  
Sony Mavica - 1981



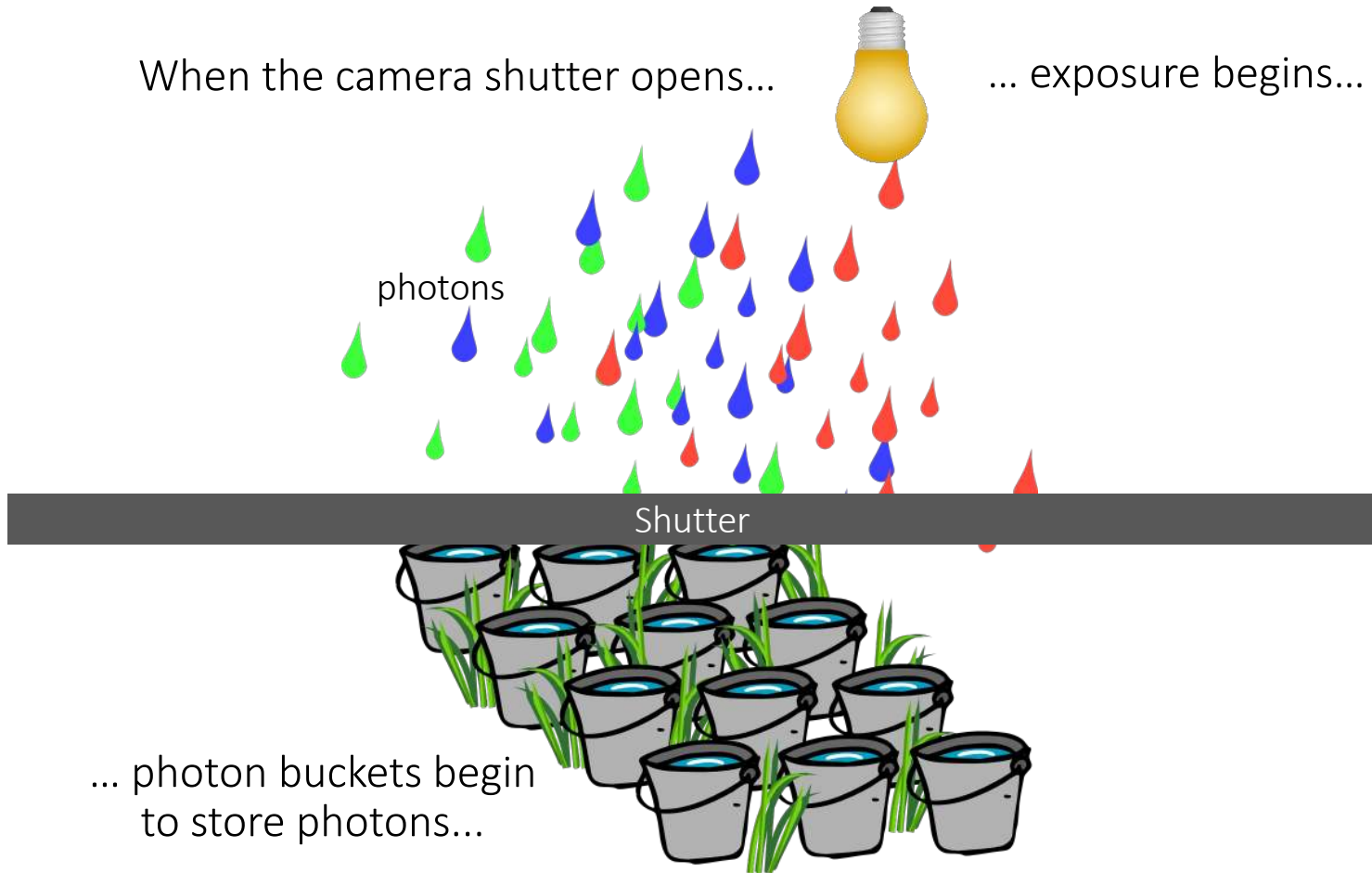
First commercially available digital camera  
Kodak DCS 100 - \$20,000 in 1991

# What does an imaging sensor do?



Canon 6D sensor  
(20.20 MP, full-frame)

# What does an imaging sensor do?



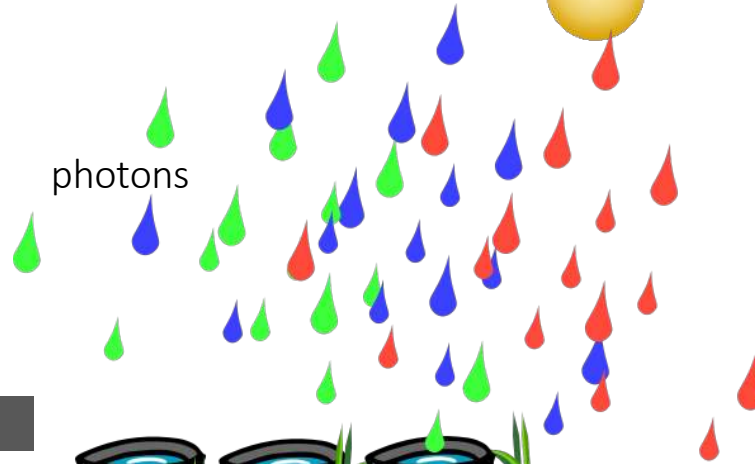
# What does an imaging sensor do?

When the camera shutter opens...



... exposure begins...

photons



... photon buckets begin  
to store photons...

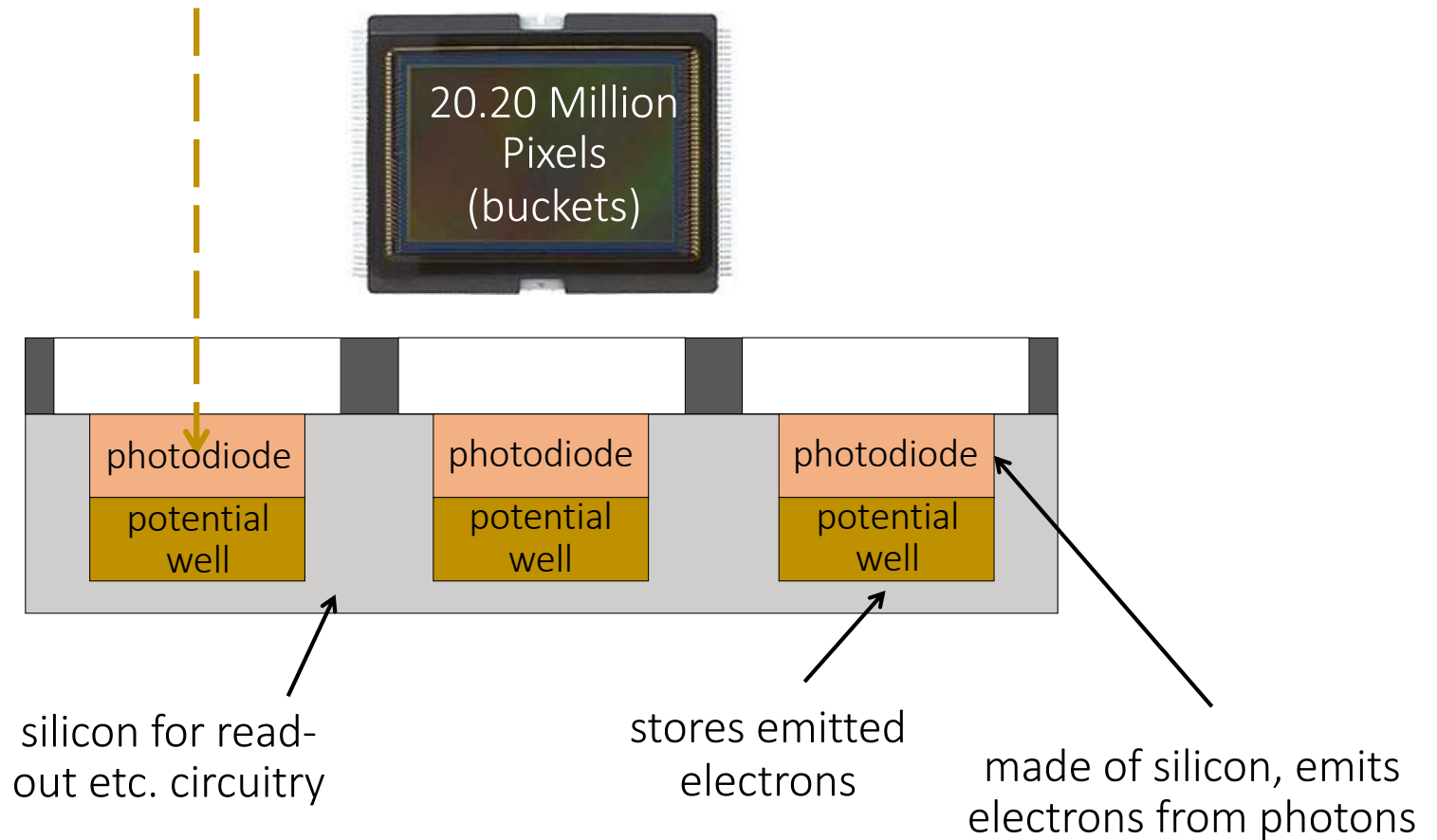


... until the camera shutter closes.  
Then, they convert stored  
photons to intensity values.

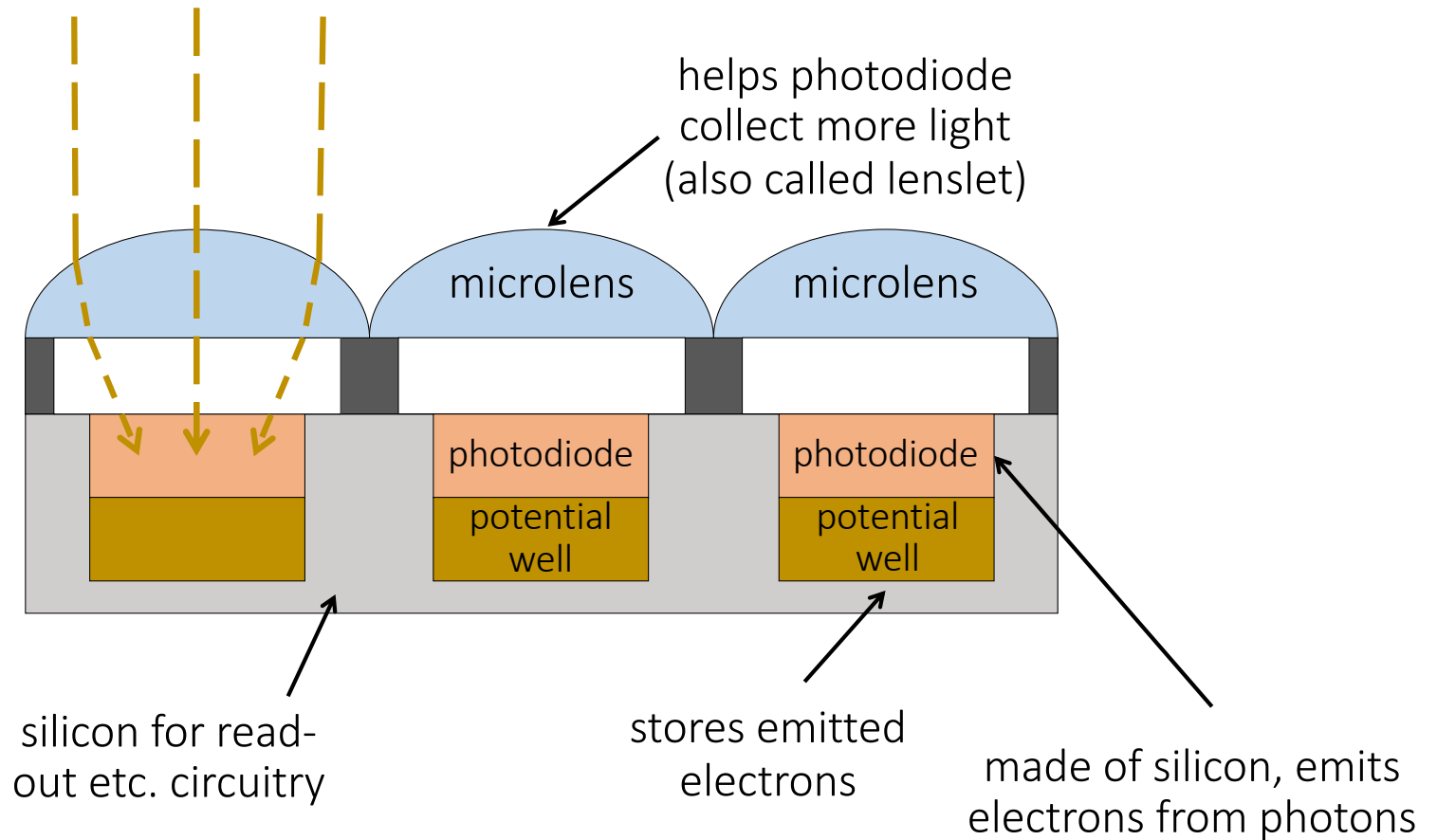


**Nikon D3s**

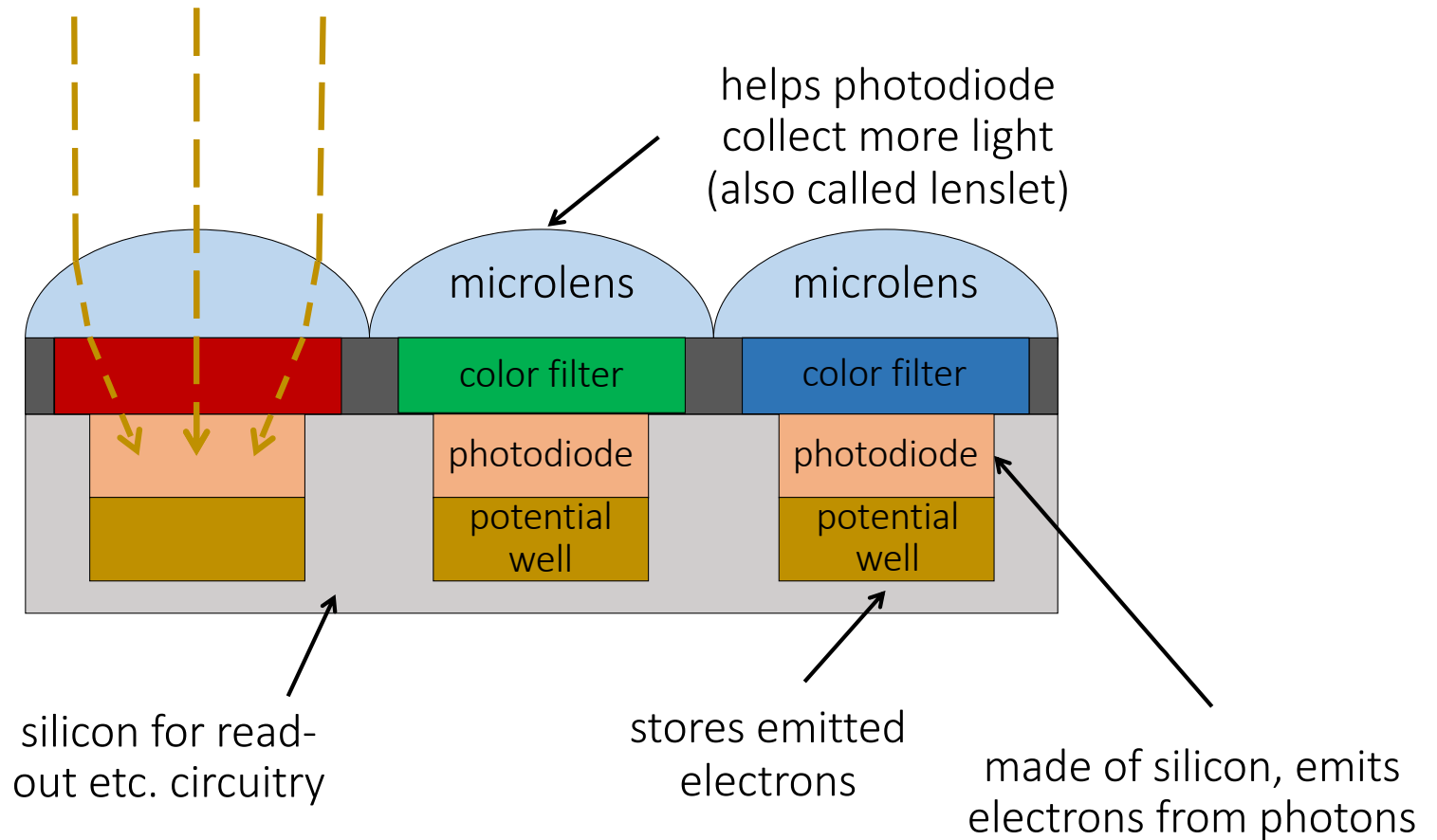
# Basic imaging sensor design



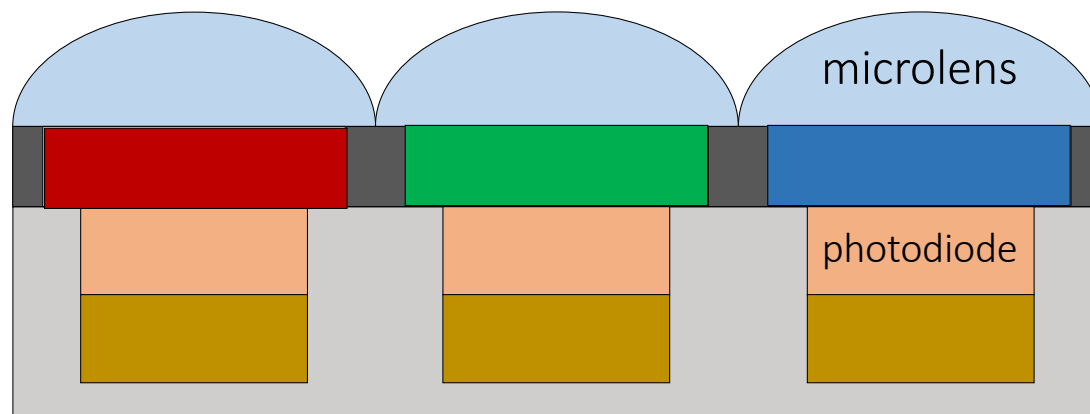
# Basic imaging sensor design



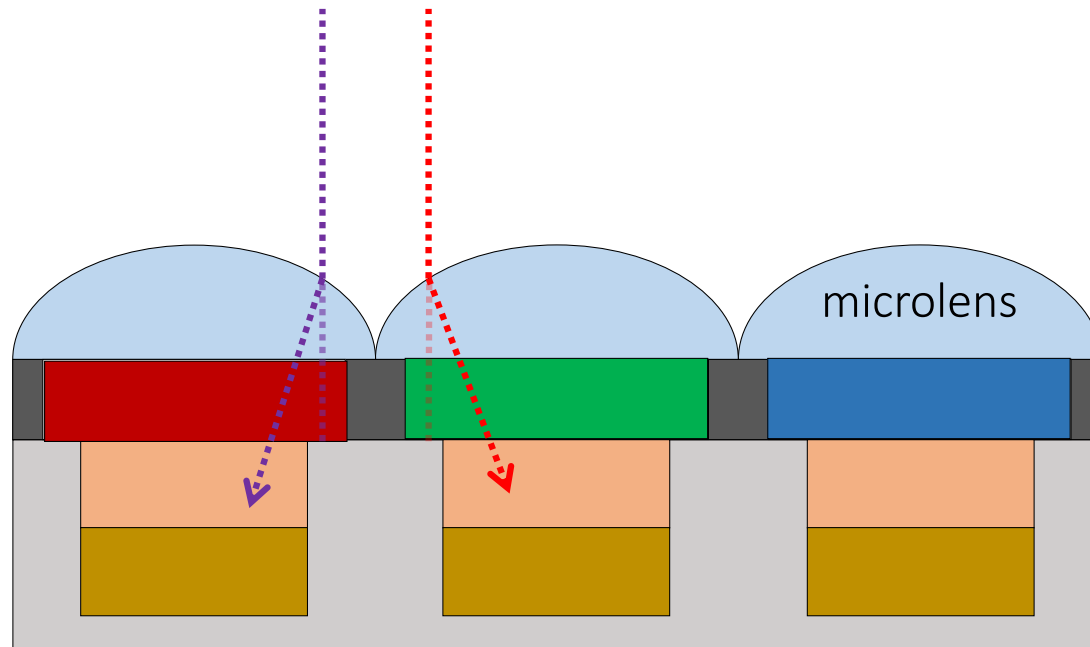
# Basic imaging sensor design



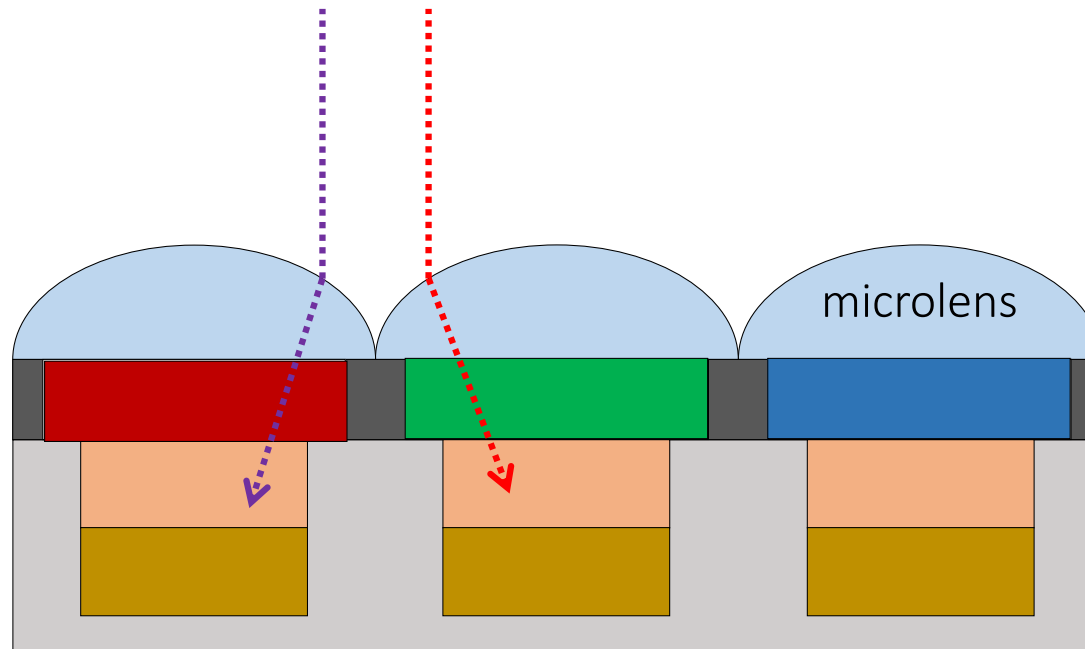
# Microlens



# Microlens



# Microlens

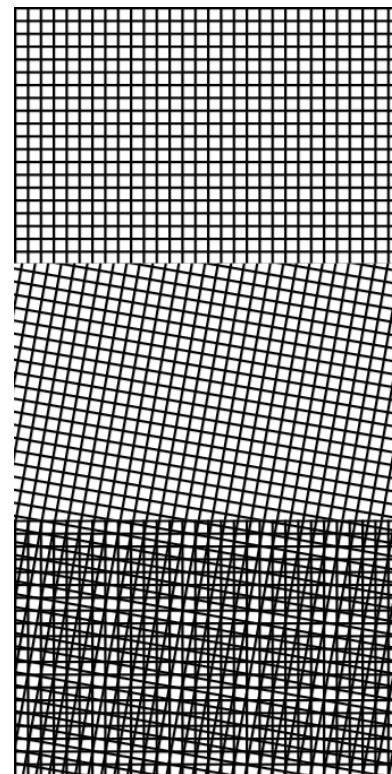
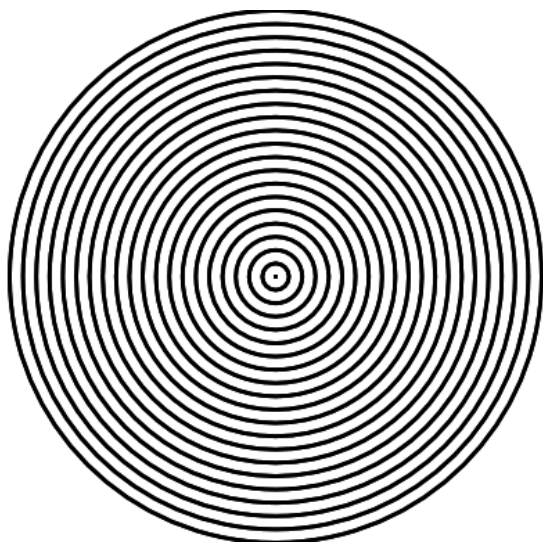


integrating light over an area at each pixel site instead of point sampling serves two functions:

- captures more photons, to improve *dynamic range*
- convolves the image with a rectangle before sampling, to avoid *aliasing*



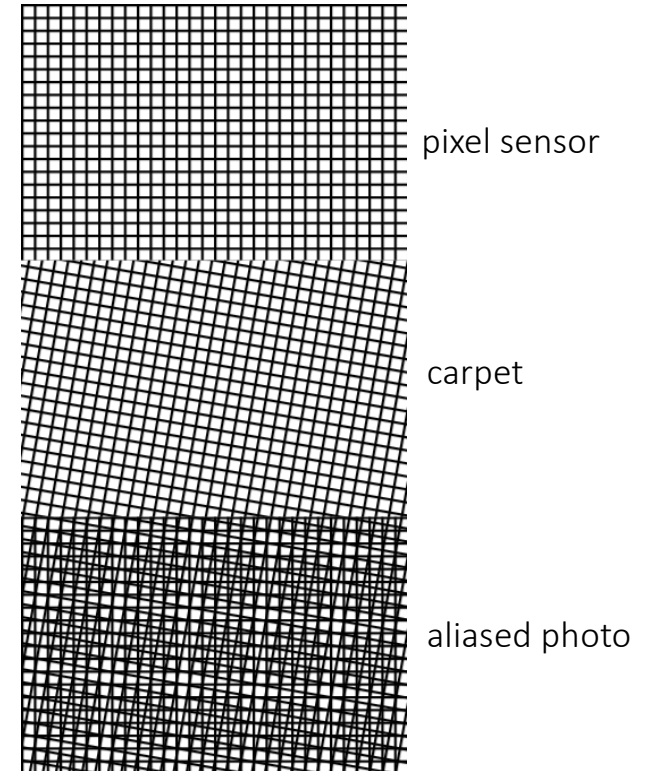
# Aliasing: Moiré Pattern



# Aliasing



caused by undersampling

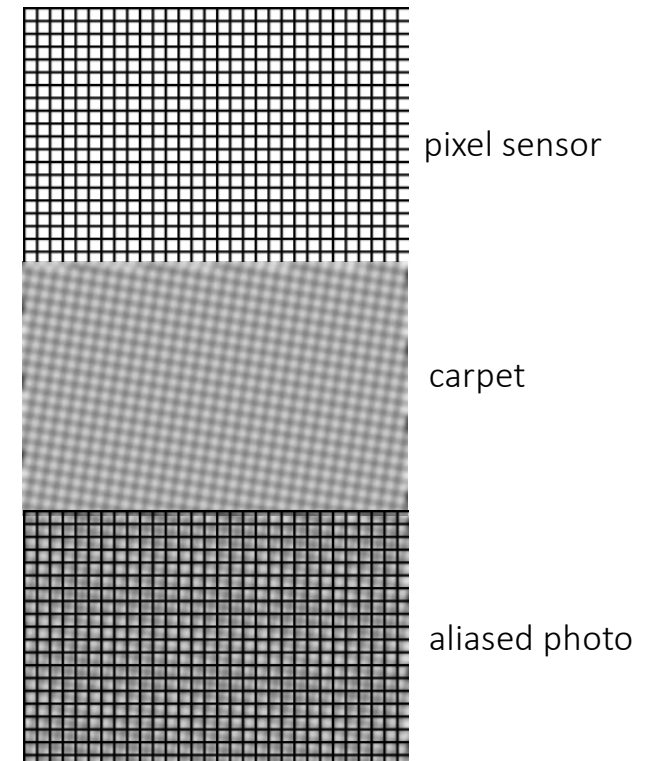


Moiré Pattern

# Aliasing



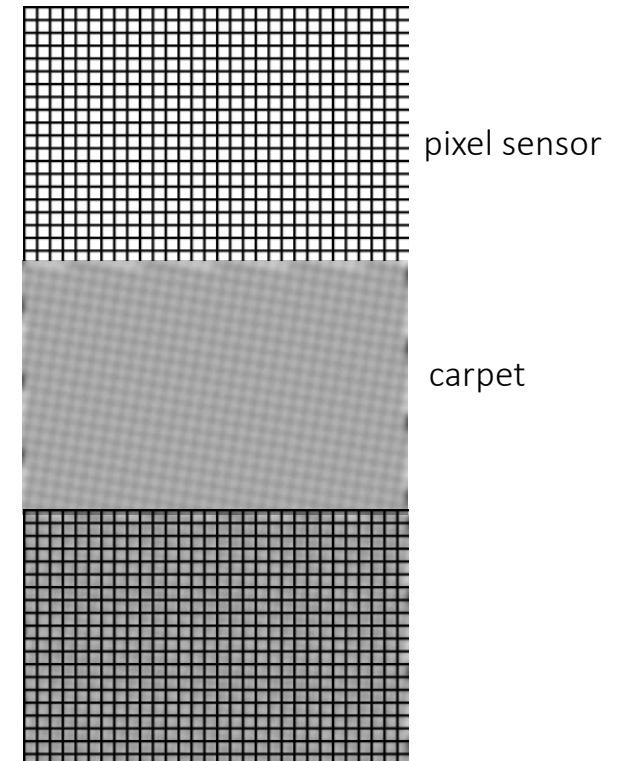
caused by undersampling  
can't correct with post processing



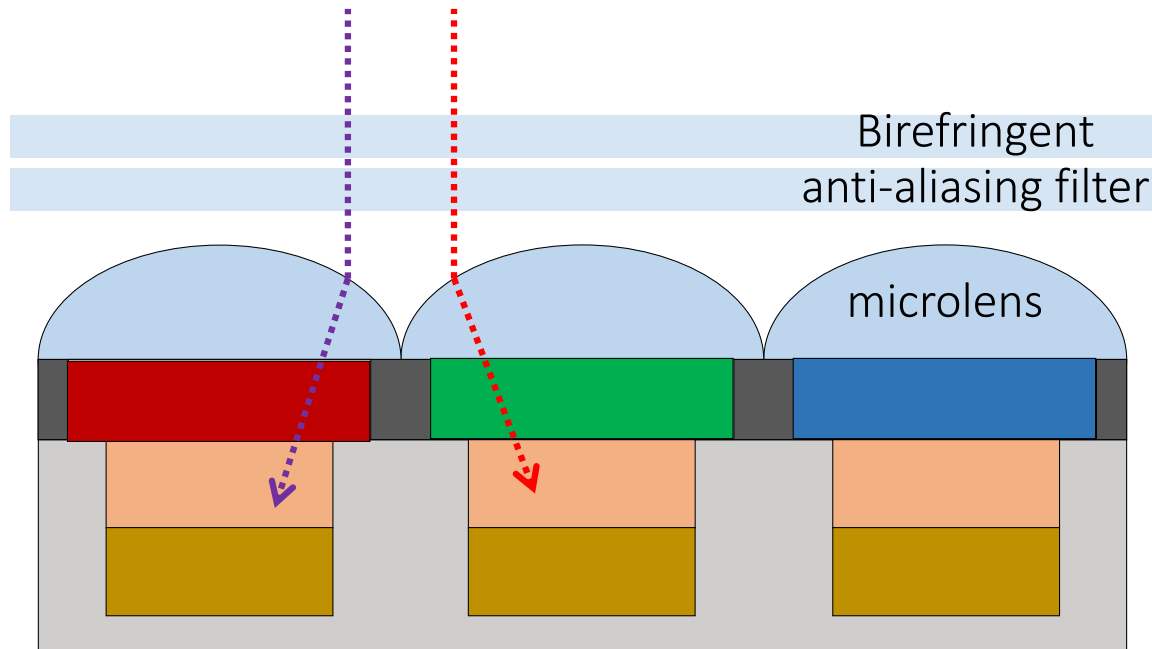
# Aliasing



caused by undersampling  
can't correct with post processing



# Optical anti-aliasing Filter

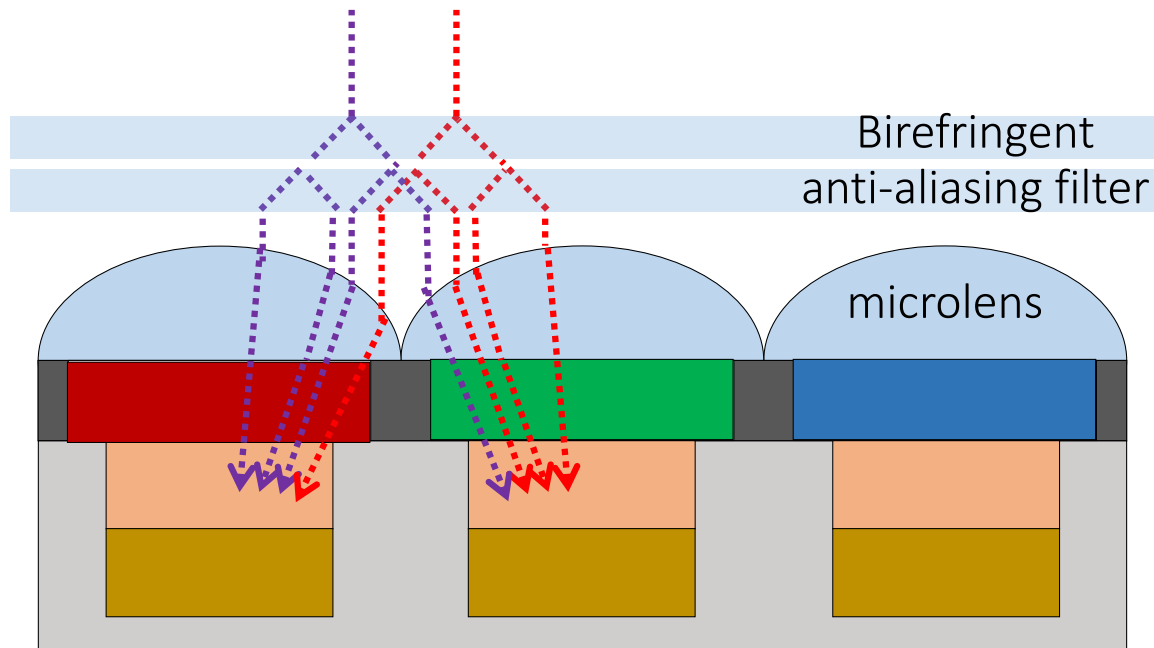


Birefringent material

Light moving through birefringent material is split into two rays taking slightly different paths (due to polarization)



# Optical anti-aliasing Filter



Birefringent material

Light moving through birefringent material is split into two rays taking slightly different paths (due to polarization)

# Optical anti-aliasing Filter

without filter



with filter



# Optical anti-aliasing Filter

without filter



with filter



What is the downside to using this filter?



# Optical anti-aliasing

without filter



with filter



the anti-aliasing filter is not always necessary, and lowers the resolution

# Optical anti-aliasing

- Nowadays, due the large number of sensor pixels, anti-aliasing filters are becoming unnecessary.



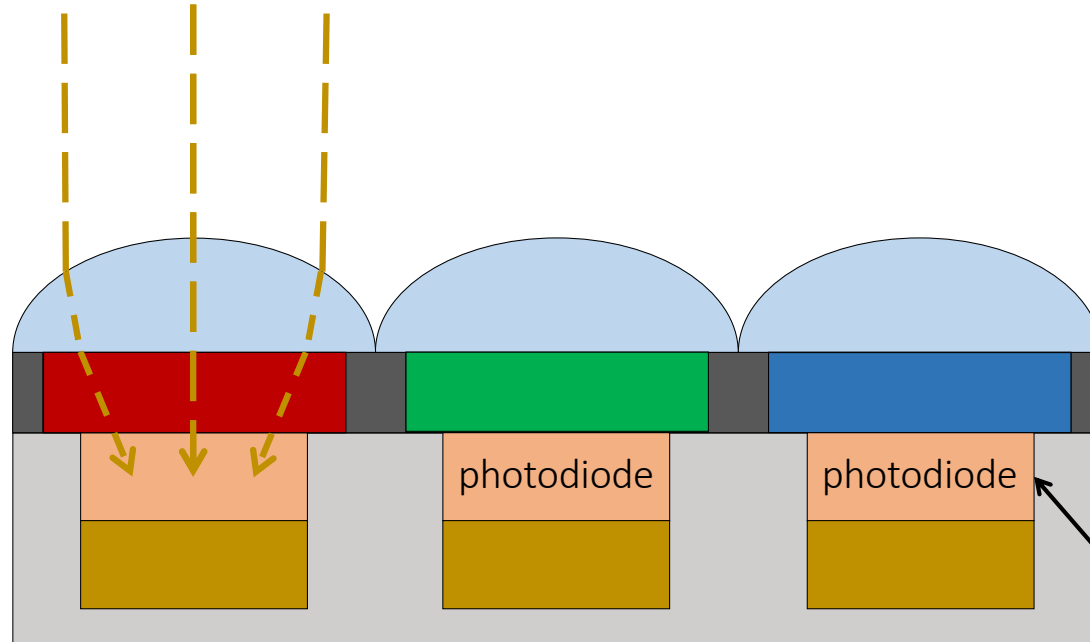
Photographers often hack their cameras to remove the AA filter, in order to avoid the loss of resolution.

a.k.a. “hot rodding”

# Sensor quality depends on (unordered)

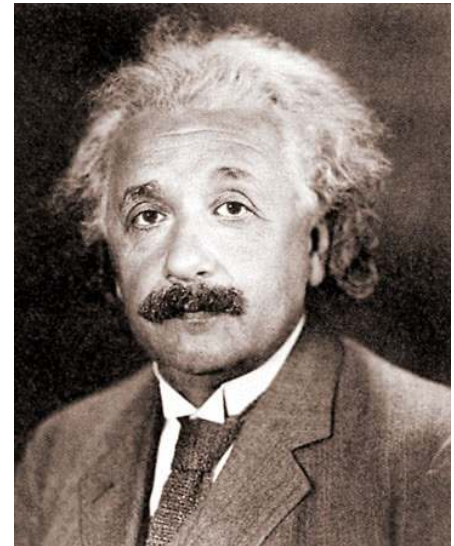
- Number of pixels (to avoid aliasing)

# Photodiode



made of silicon, emits  
electrons from photons

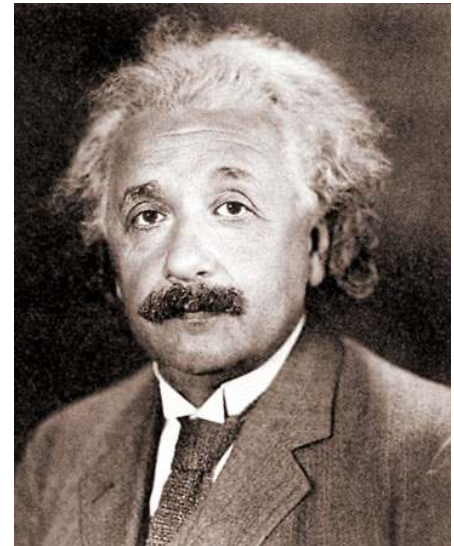
# Nobel Prize in Physics



Albert Einstein

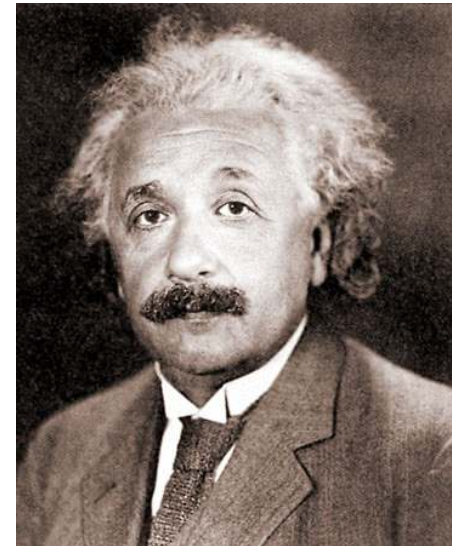
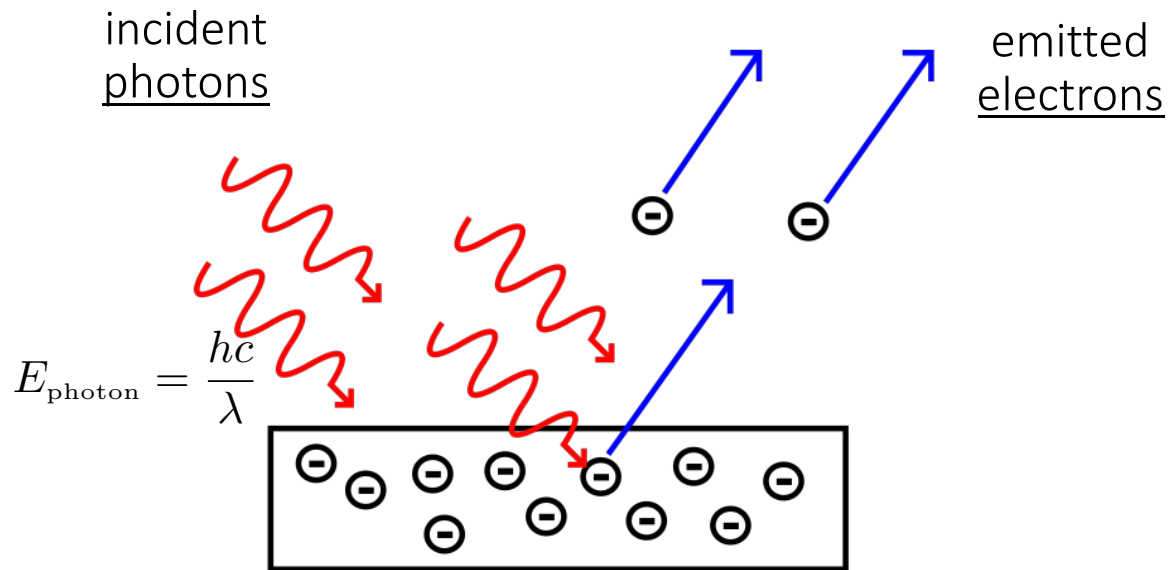
# Nobel Prize in Physics

What is he known for?



Albert Einstein

# Photoelectric effect

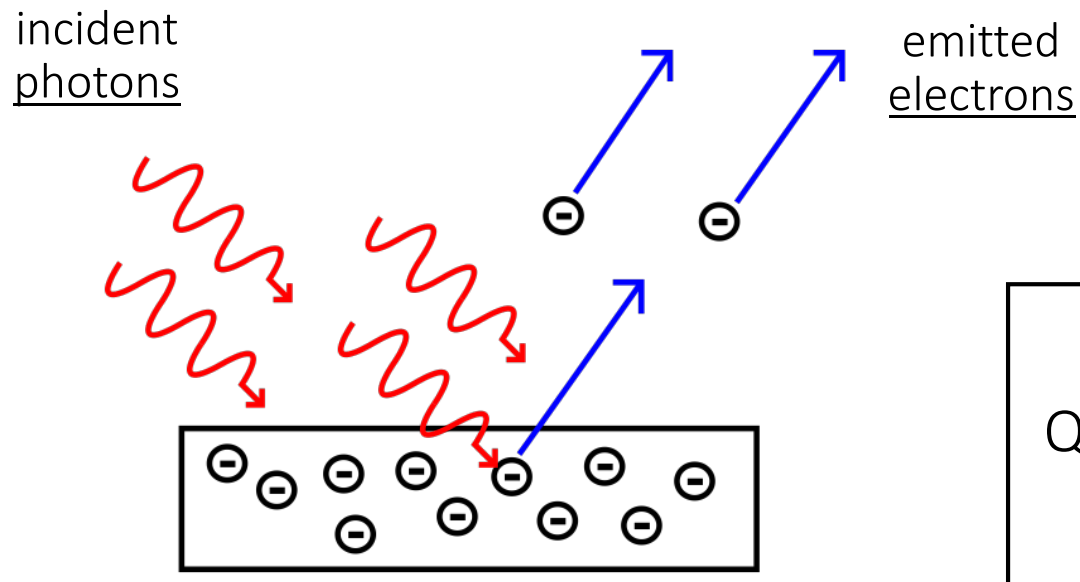


Albert Einstein

Einstein's Nobel Prize in 1921 "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect"



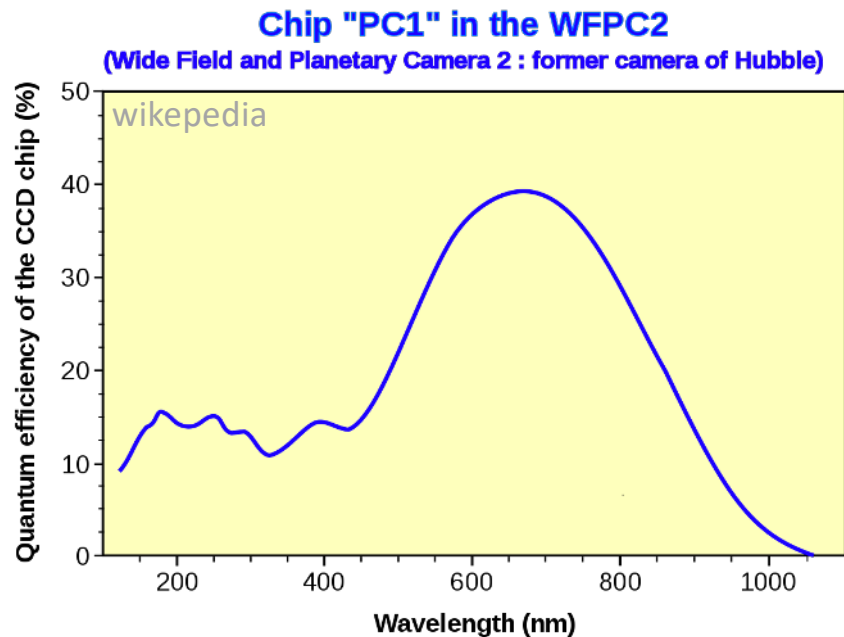
# Photodiode quantum efficiency (QE)



$$QE = \frac{\text{\# electrons}}{\text{\# photons}}$$

How many of the incident photons will the photodiode convert into electrons?

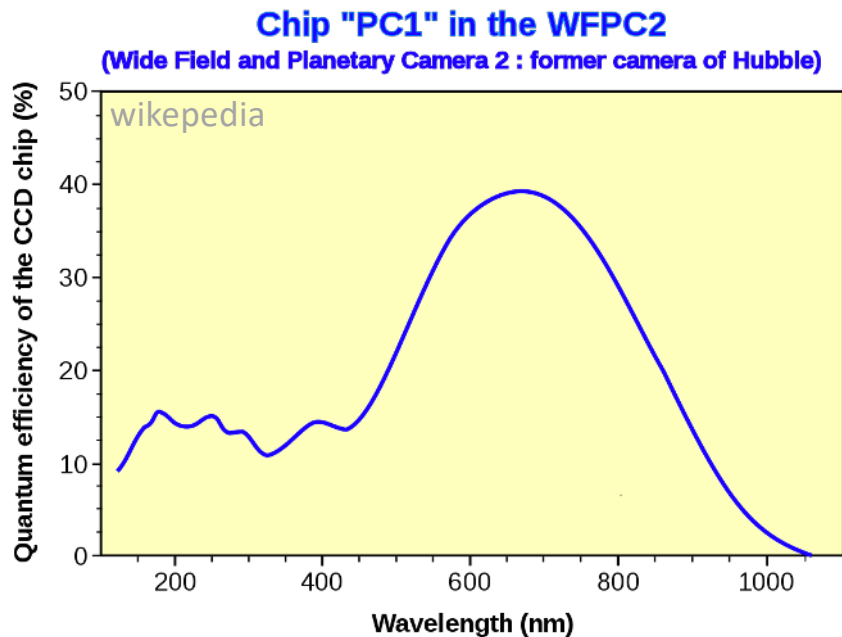
# Photodiode quantum efficiency (QE)



← Hubble Camera QE

$$QE = \frac{\text{\# electrons}}{\text{\# photons}}$$

# Photodiode quantum efficiency (QE)



← Hubble Camera QE

$$QE = \frac{\text{\# electrons}}{\text{\# photons}}$$

If QE is larger then is the camera better or worse at that wavelength?

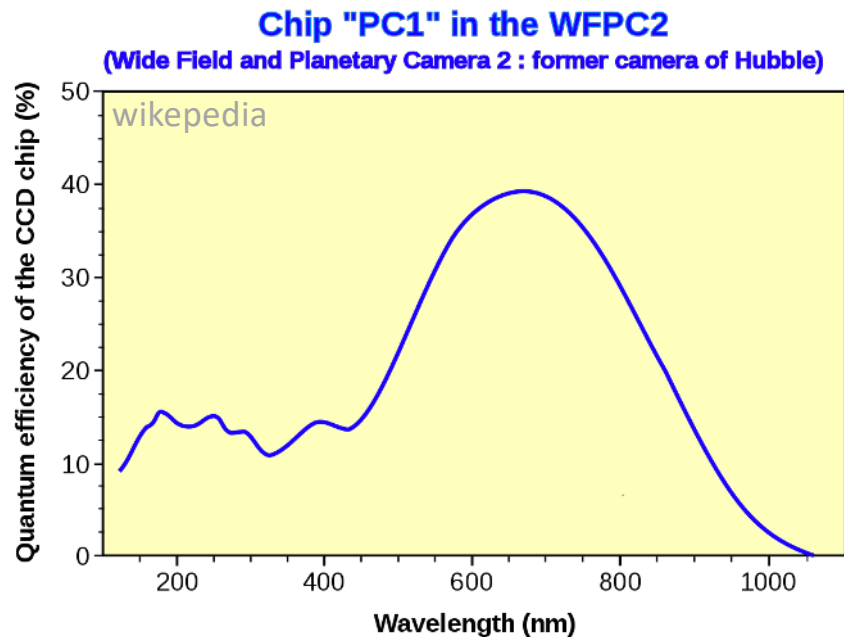


worse



better

# Photodiode quantum efficiency (QE)



← Hubble Camera QE

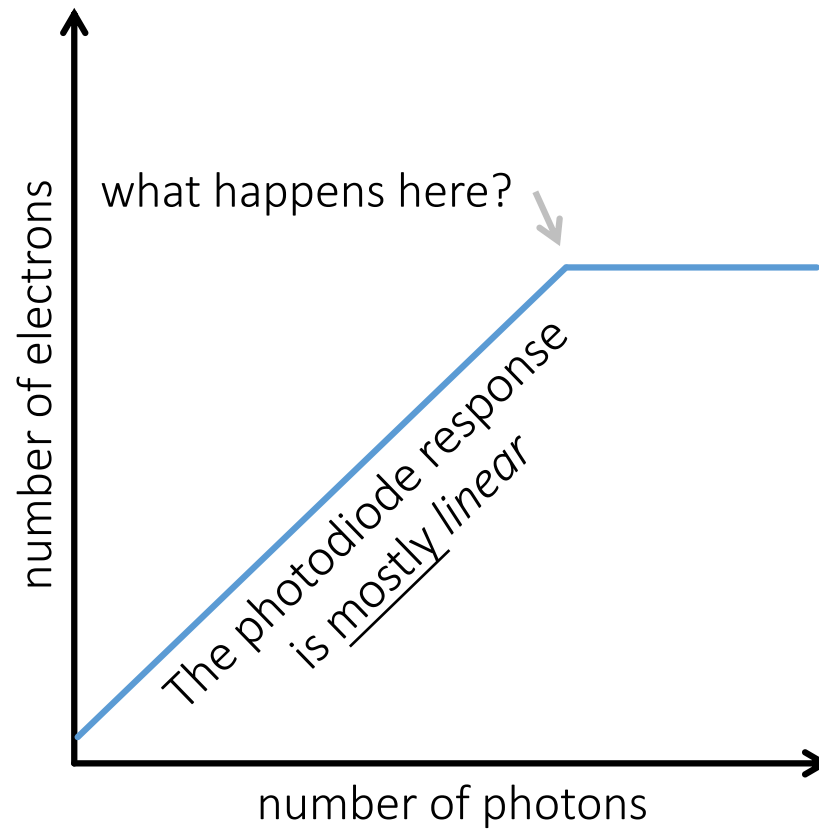
$$QE = \frac{\text{\# electrons}}{\text{\# photons}}$$

Human vision: ~15%    Typical digital camera: <50%    Best CCD cameras: ~90%

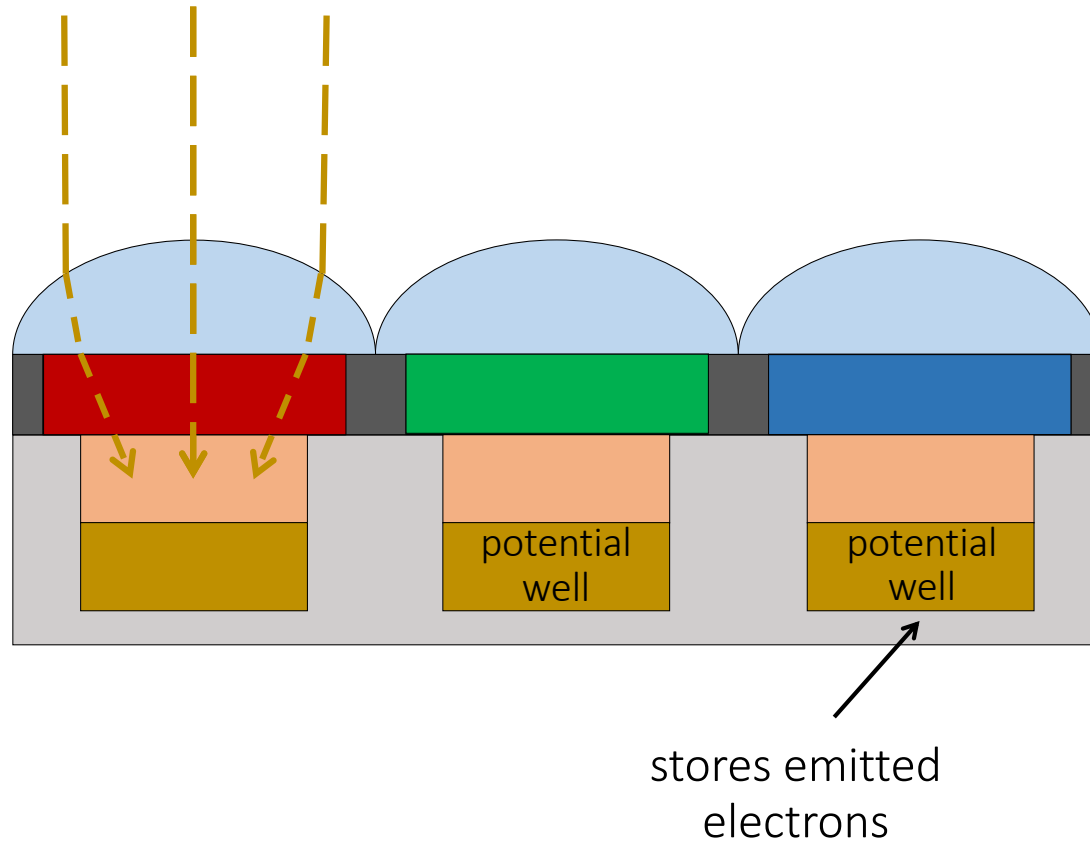
# Sensor quality depends on (unordered)

- Number of pixels (to avoid aliasing)
- Quantum efficiency (QE) to capture the most # of photons

# Photodiode response

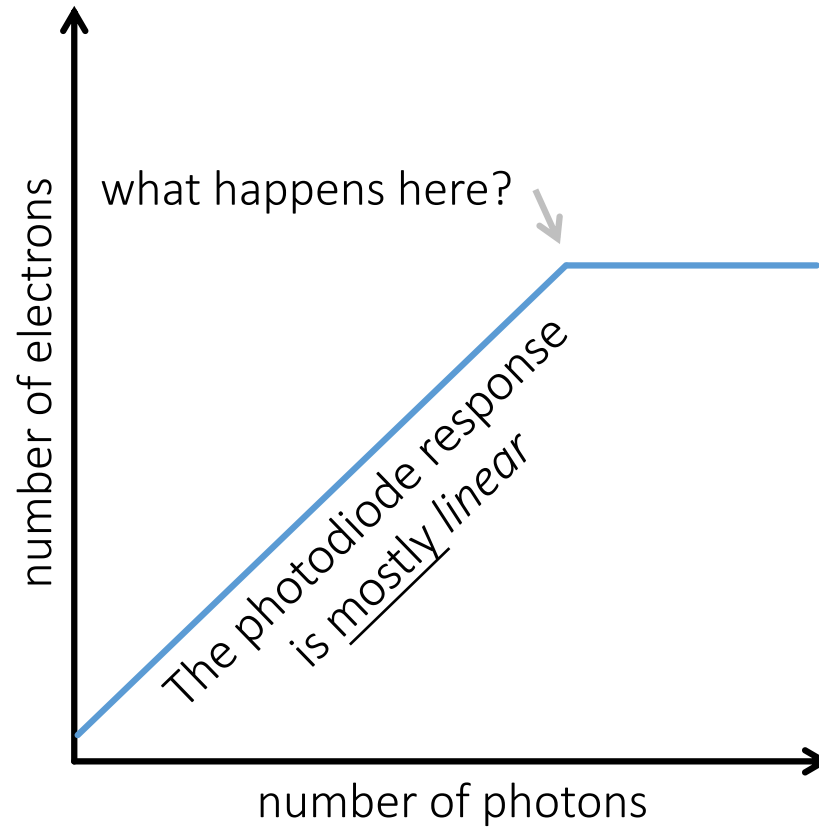


# Photodiode





# Photodiode response



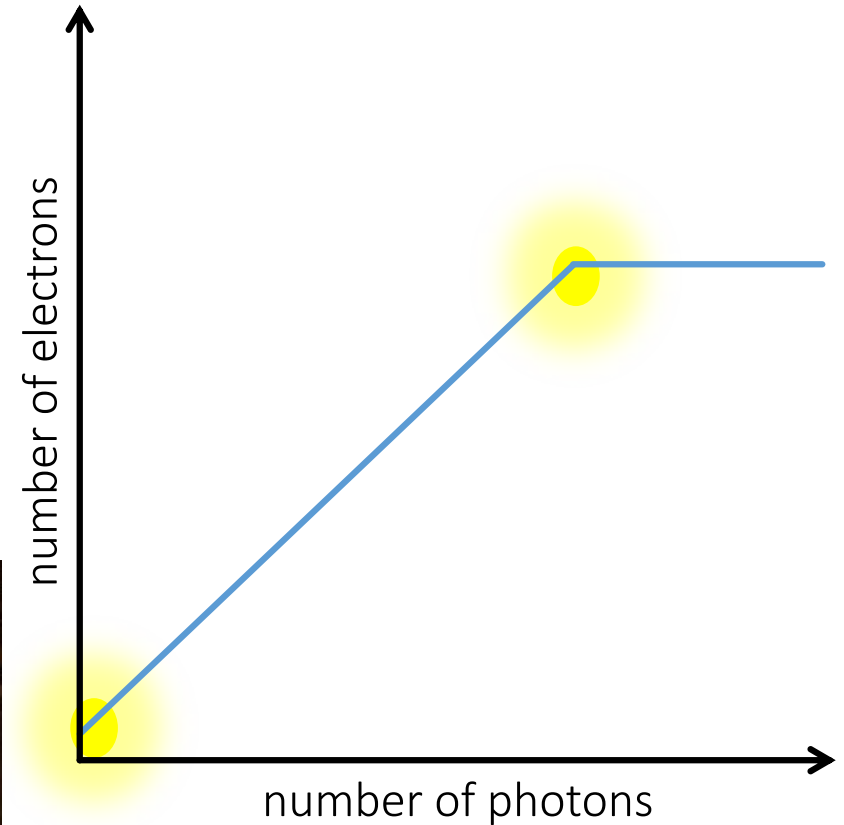
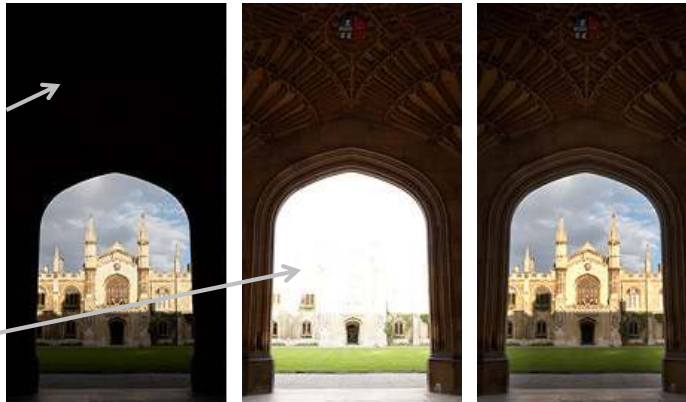
# Photodiode response

- non-linear when potential well is full before exposure ends (over-exposure)
- non-linear near zero (due to noise)

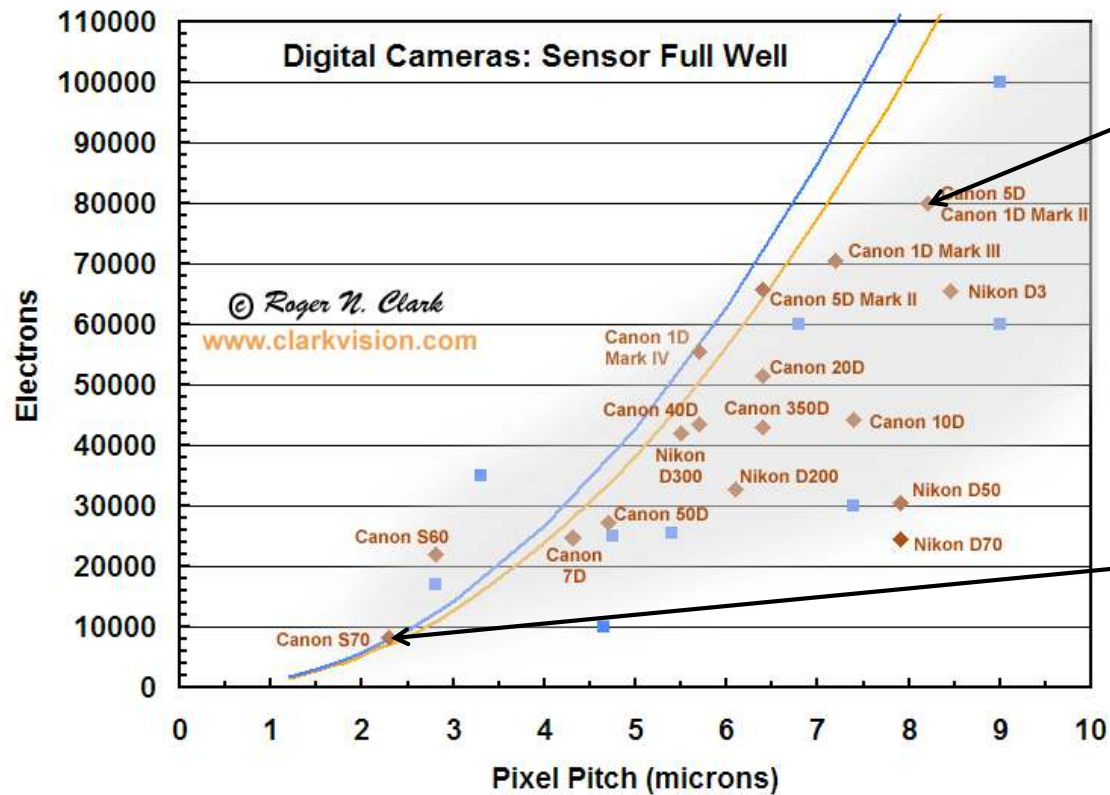
We will see how to deal with these issues in a later lecture (high-dynamic-range imaging).

under-exposure  
(non-linearity due  
to sensor noise)

over-exposure  
(non-linearity due  
to sensor saturation)



# Photodiode full-well capacity



How many electrons can the photodiode store before saturation?

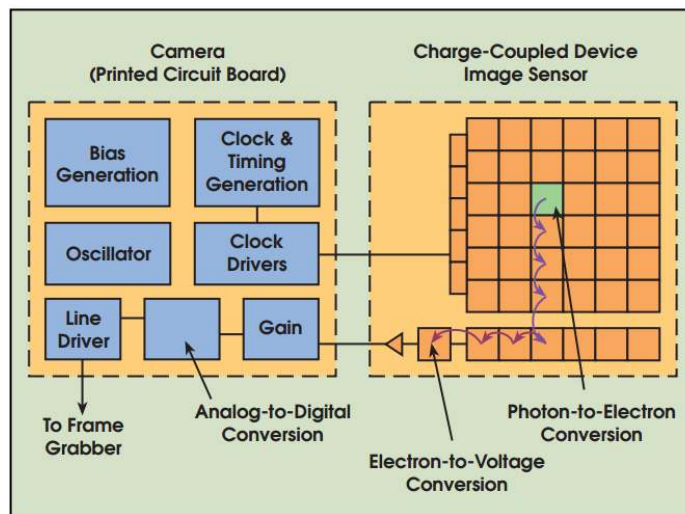
# Sensor quality depends on (unordered)

- Number of pixels (to avoid aliasing)
- Quantum efficiency (QE) to capture the most # of photons
- Photodiode's well capacity to store the most number of photons

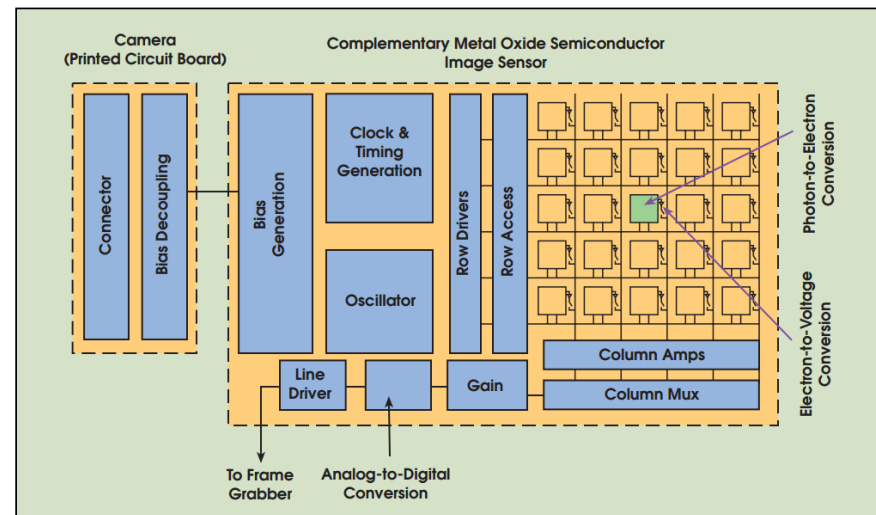
# Two main types of imaging sensors

Do you know them?

# Two main types of imaging sensors

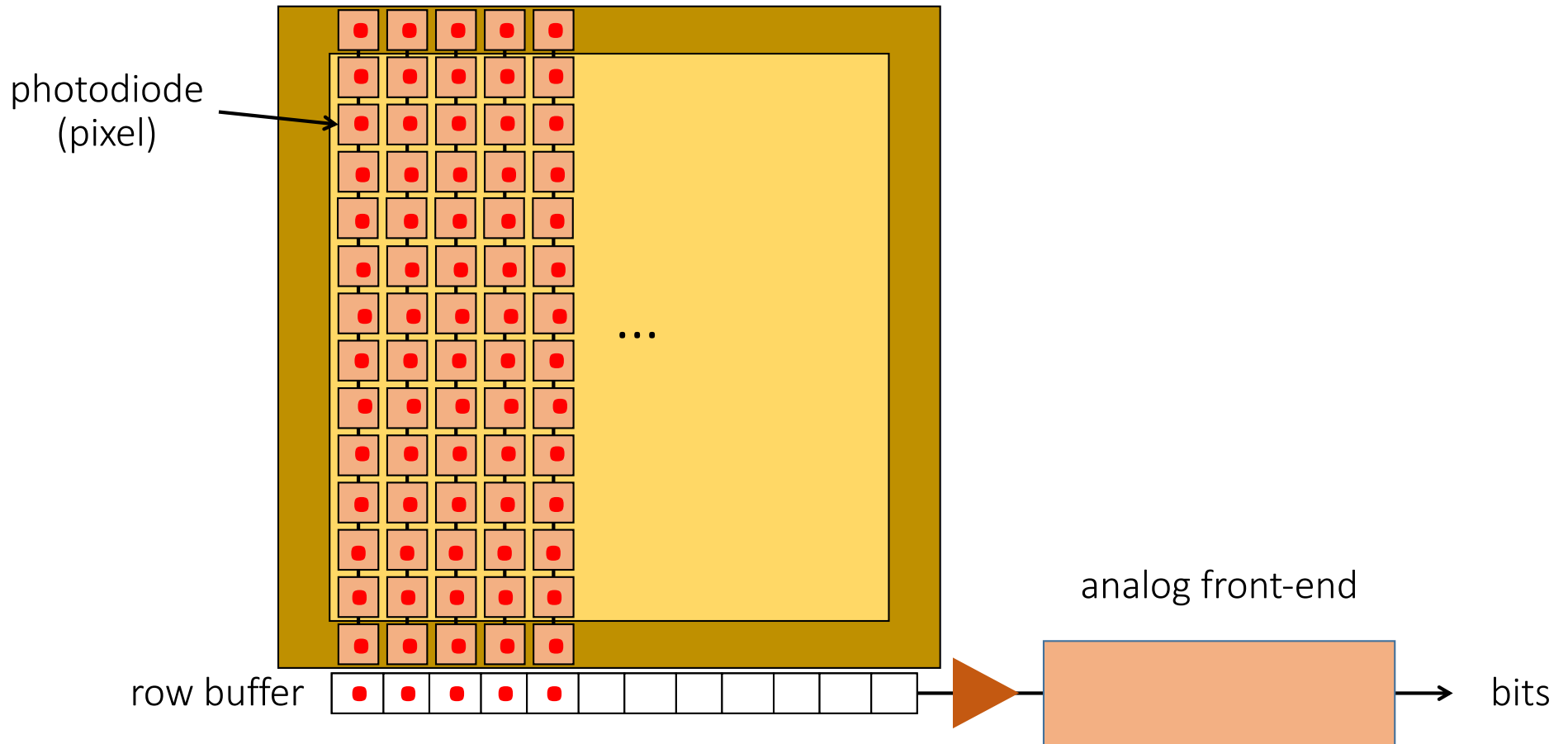


Charged Coupled Device (CCD):  
converts electrons to voltage using  
readout circuitry separate from pixel

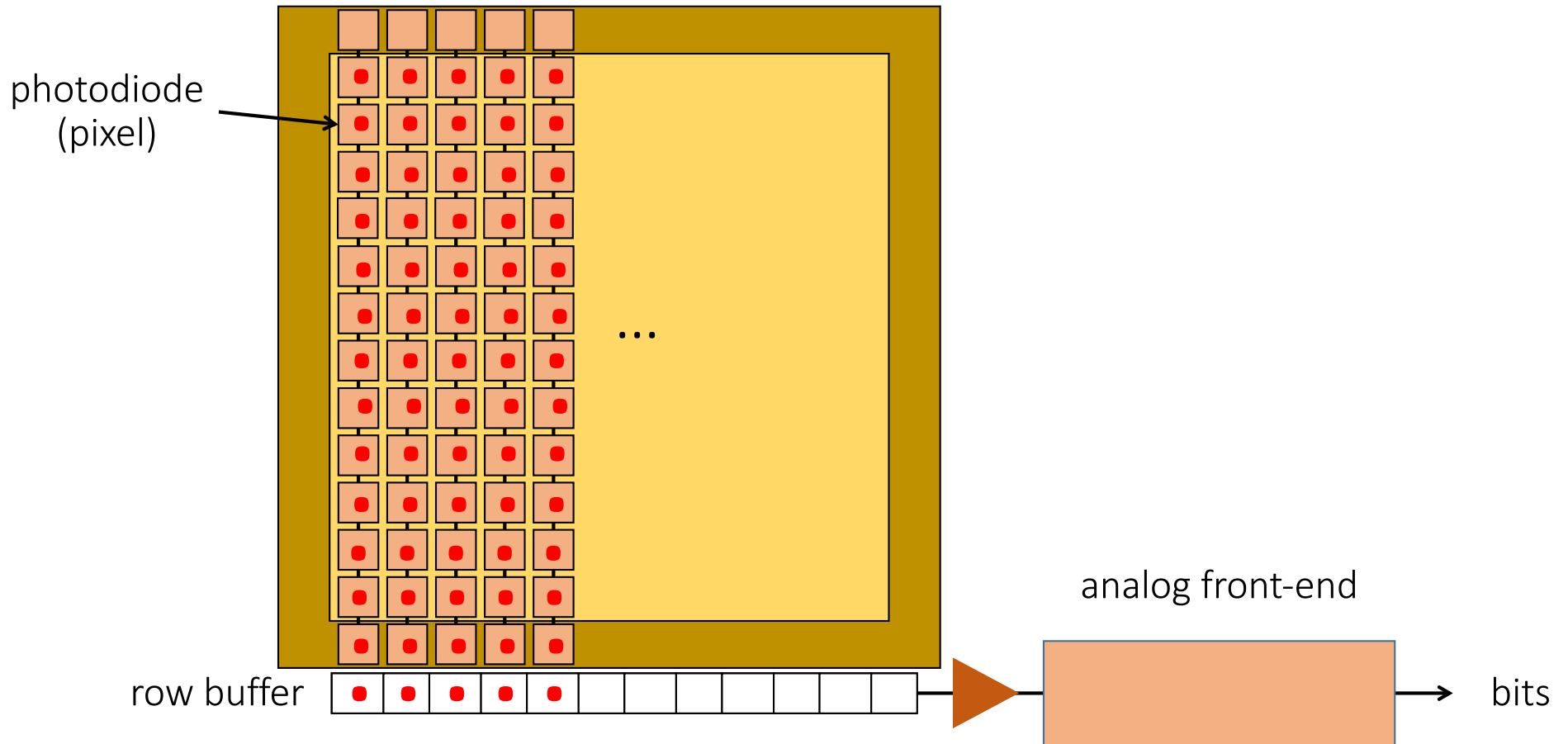


Complementary Metal Oxide Semiconductor (CMOS):  
converts electrons to voltage using  
per-pixel readout circuitry

# CCD sensor (very) simplified layout

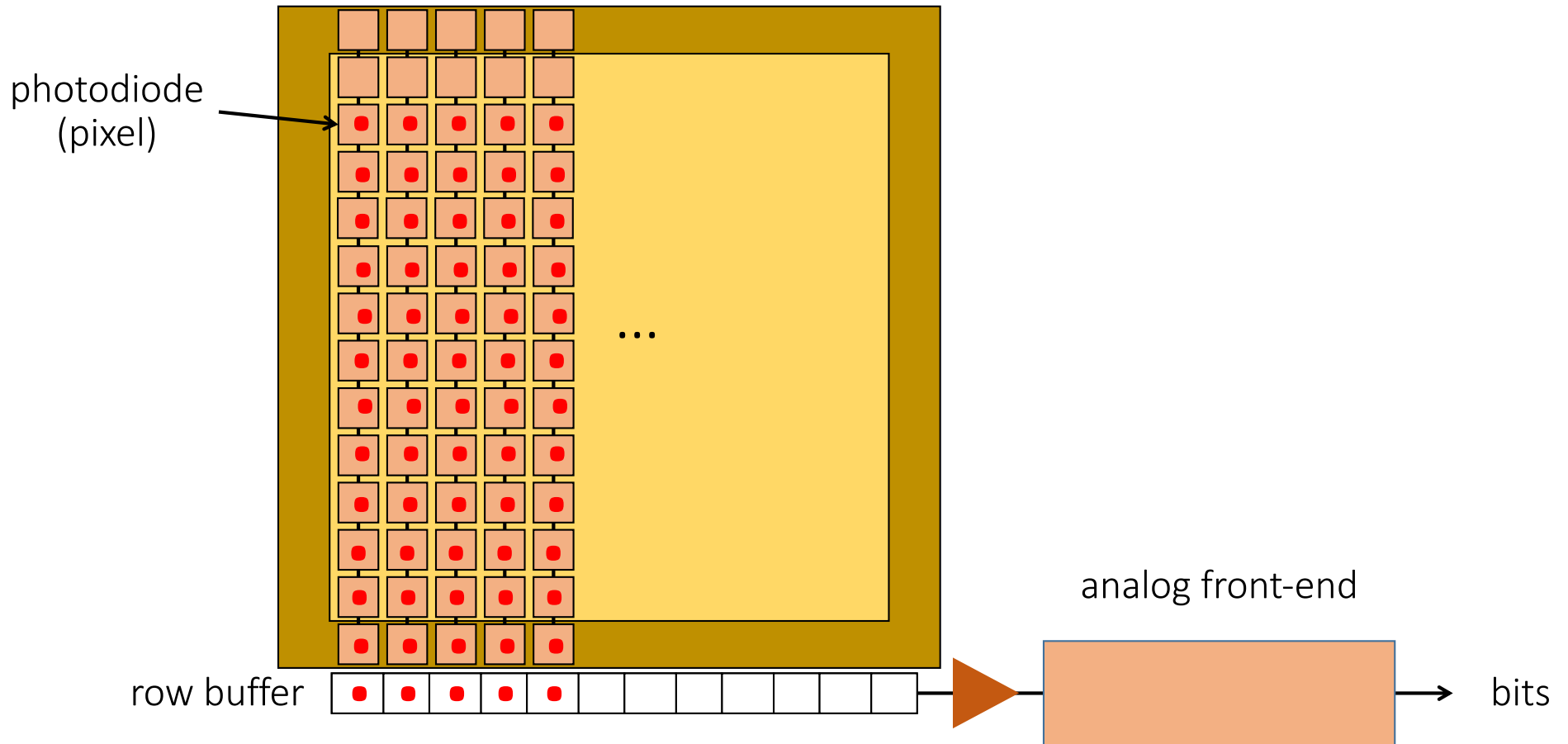


# CCD sensor (very) simplified layout

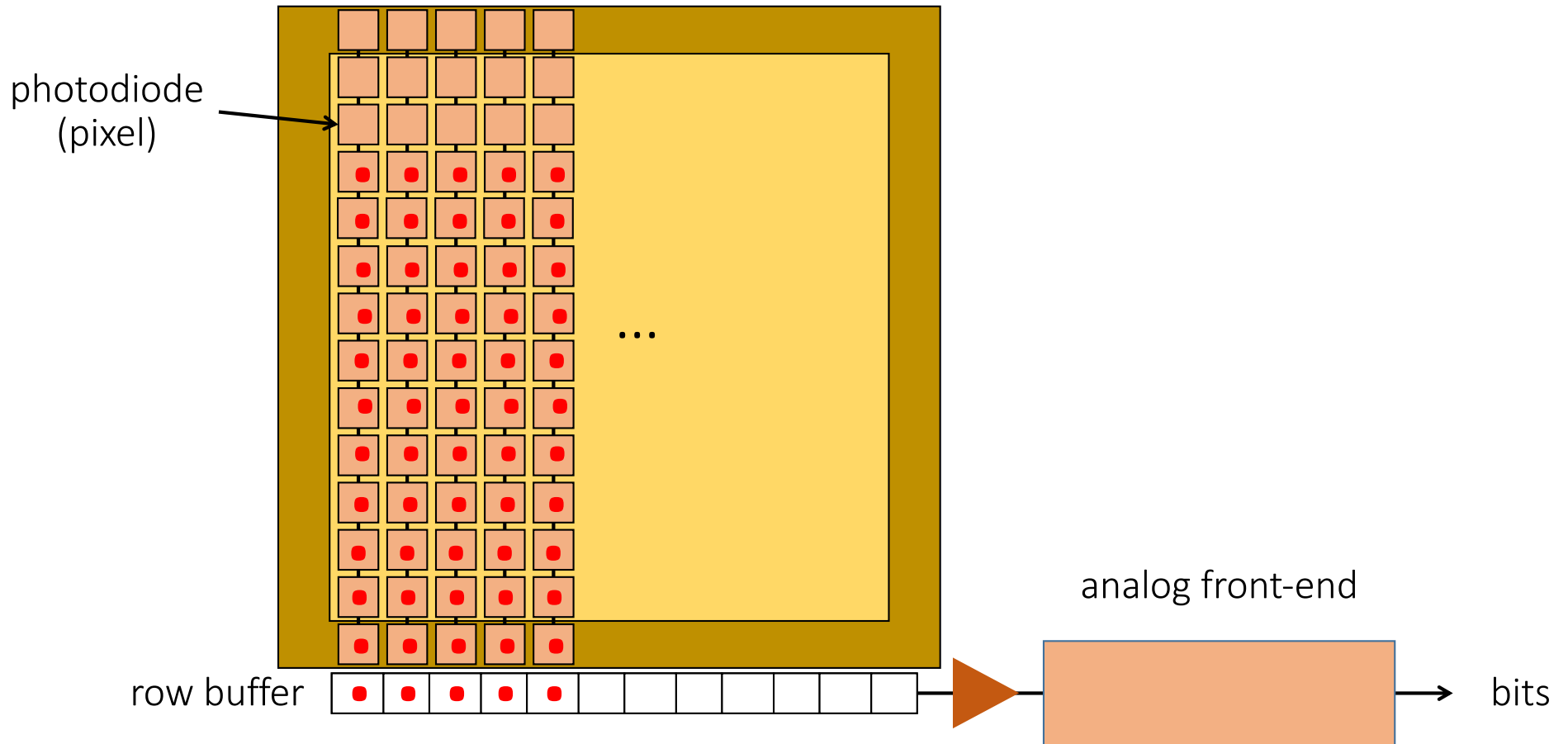




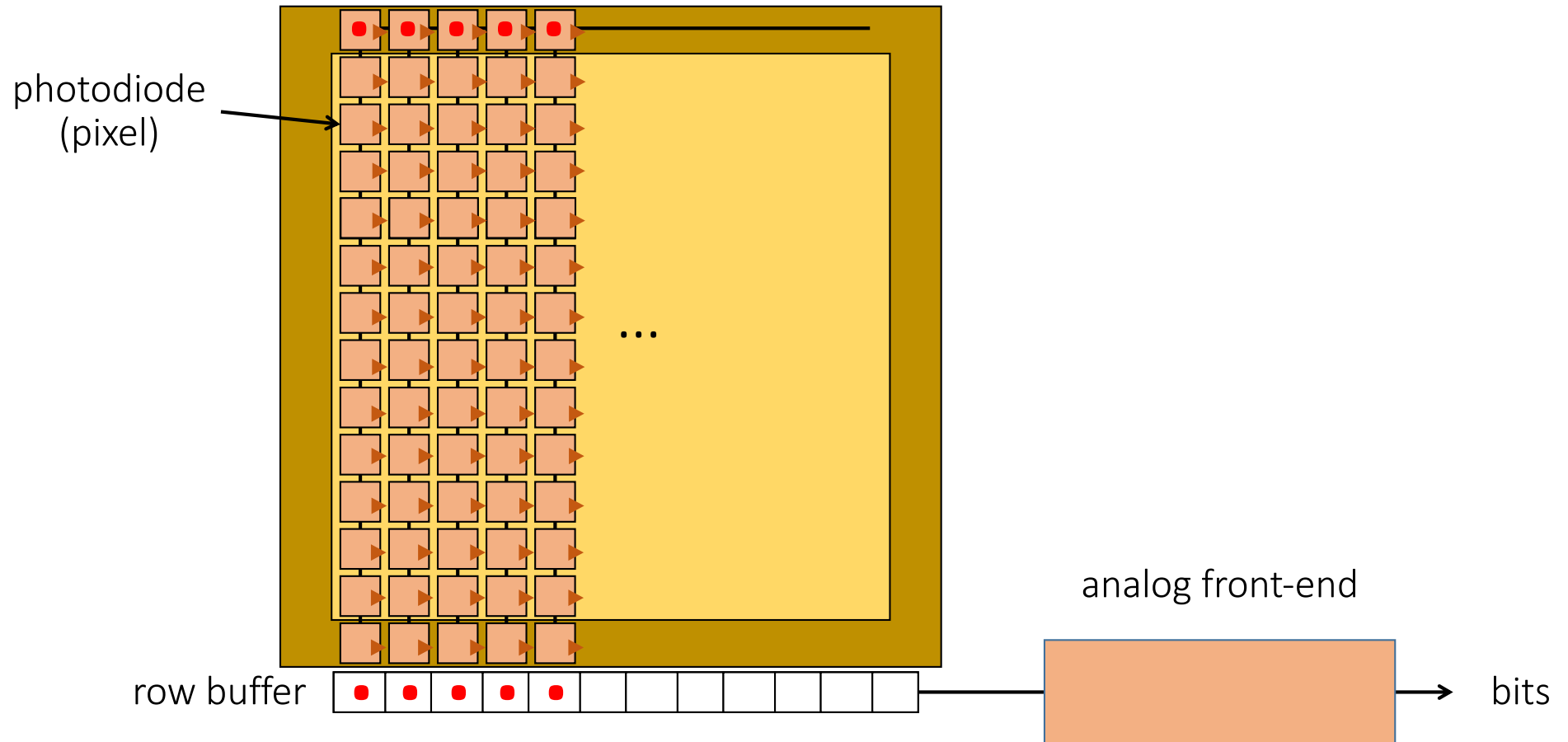
# CCD sensor (very) simplified layout



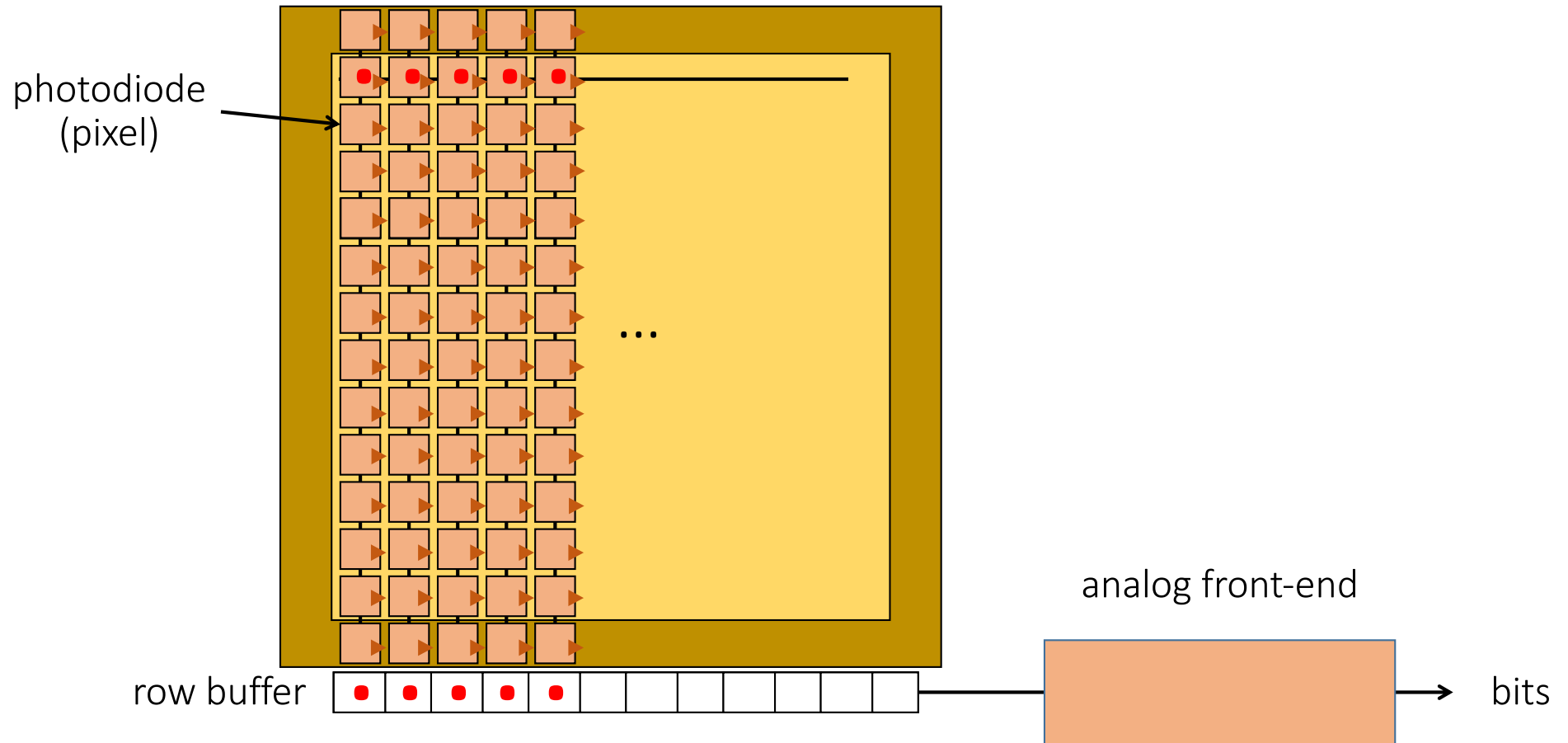
# CCD sensor (very) simplified layout



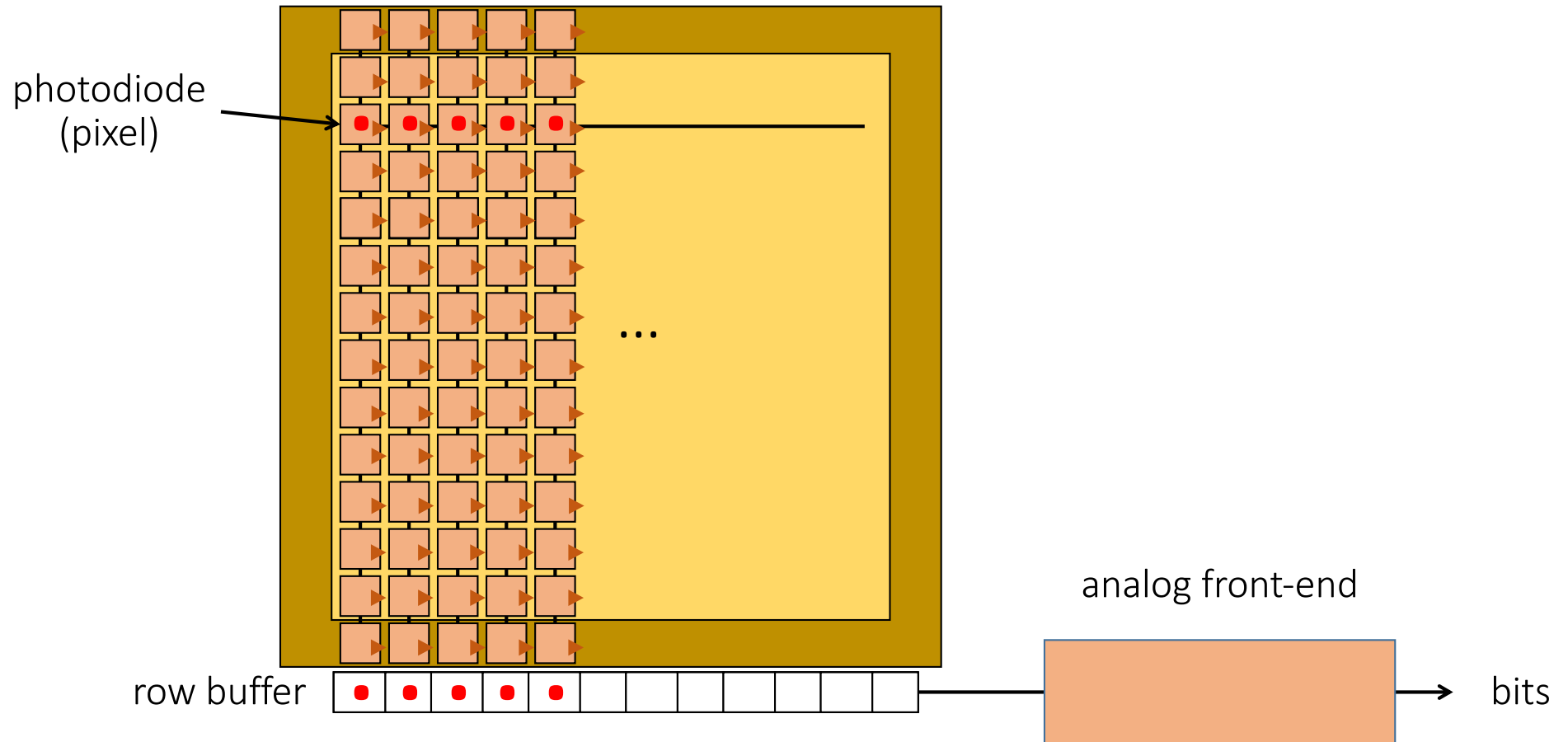
# CMOS sensor (very) simplified layout



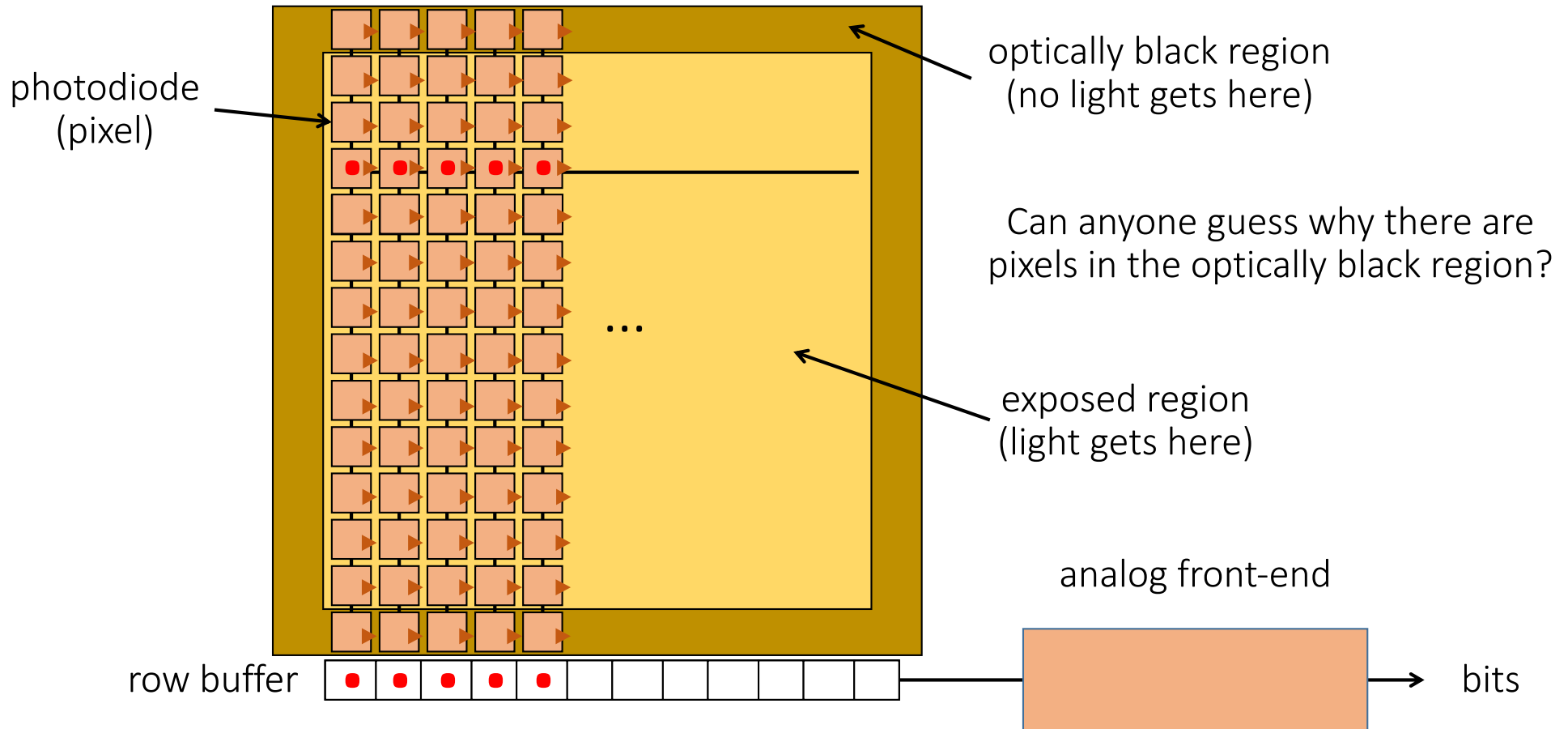
# CMOS sensor (very) simplified layout



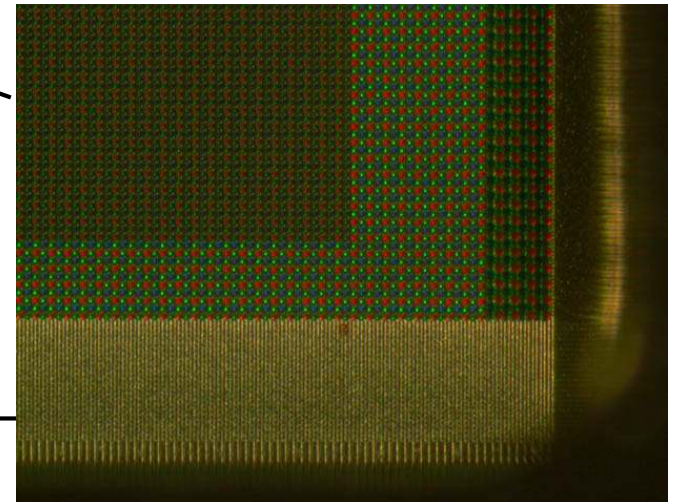
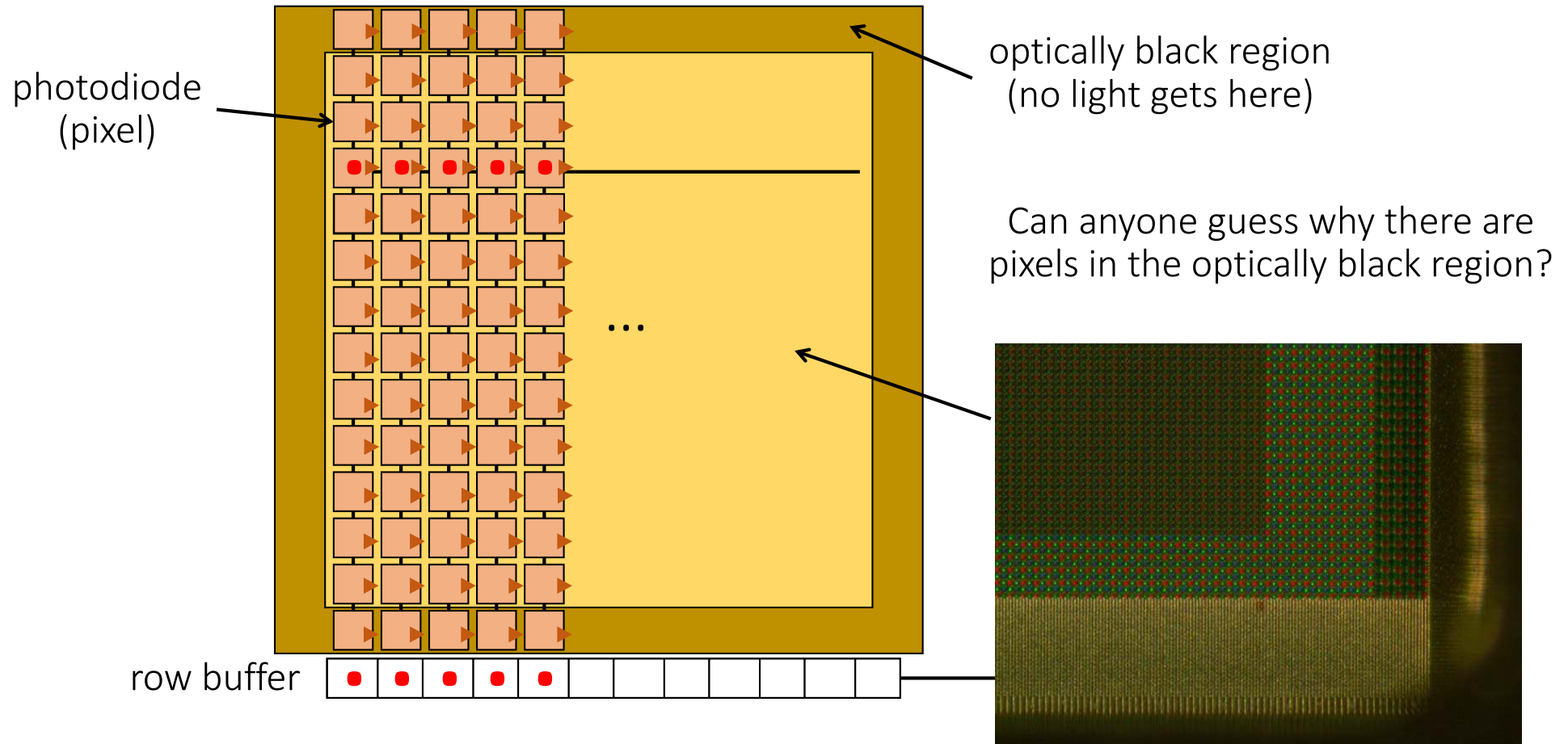
# CMOS sensor (very) simplified layout



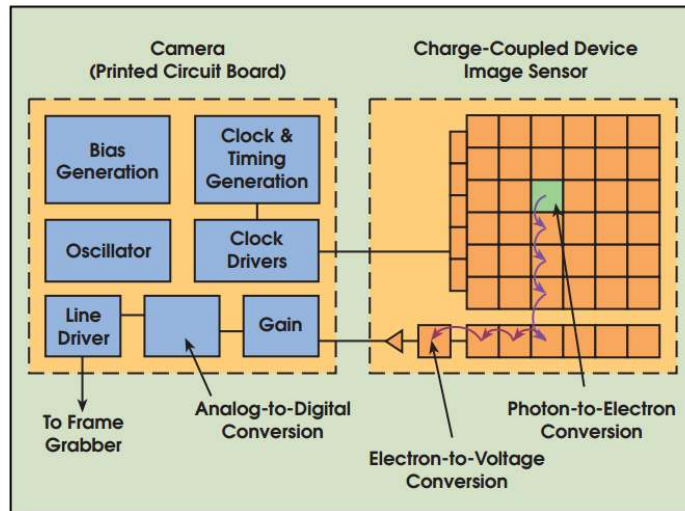
# CMOS sensor (very) simplified layout



# CMOS sensor (very) simplified layout

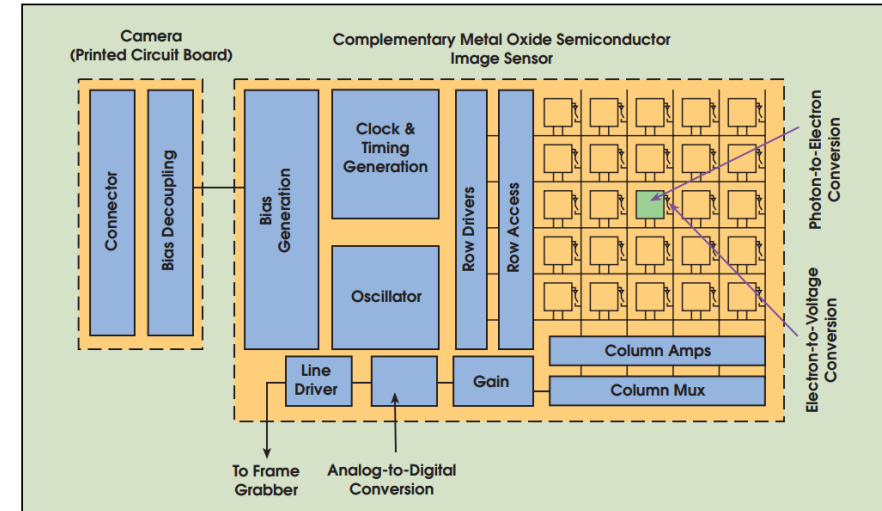
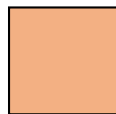


# Advantages of each sensor



Charged Coupled Device (CCD):  
converts electrons to voltage using  
readout circuitry separate from pixel

- ✓ higher sensitivity
- ✓ lower noise



Complementary Metal Oxide Semiconductor (CMOS):  
converts electrons to voltage using  
per-pixel readout circuitry

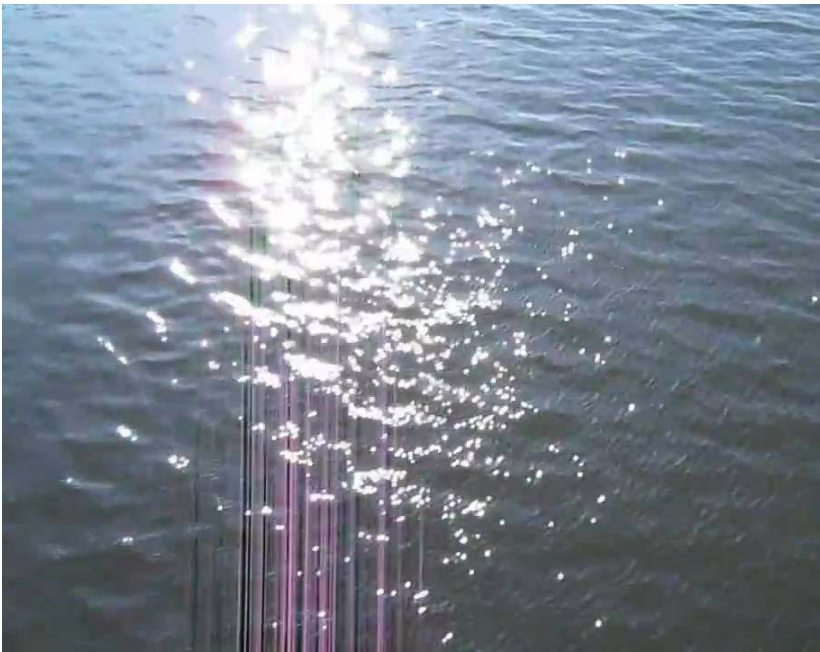
- ✓ faster read-out
- ✓ lower cost





# Common image distortions

CCD



Smearing

CMOS



Distortion due to  
Rolling shutter

# What does an imaging sensor do?

When the camera shutter opens, the sensor:

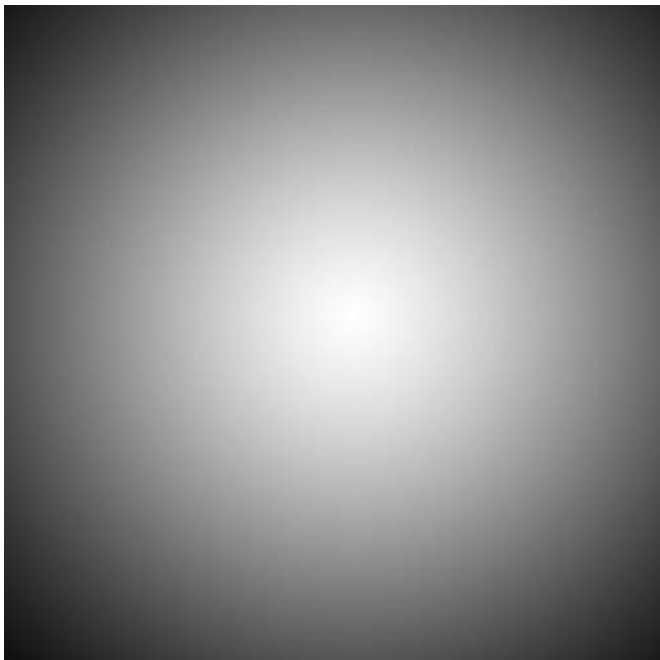
- at every photodiode, converts incident photons into electrons
- stores electrons into the photodiode's potential well while it is not full

... until camera shutter closes. Then, the analog front-end:

- reads out photodiodes' wells, row-by-row, and converts them to analog signals
- applies a (possibly non-uniform) gain to these analog signals
- converts them to digital signals

# Vignetting

Fancy word for: pixels far off the center receive less light



white wall under uniform light

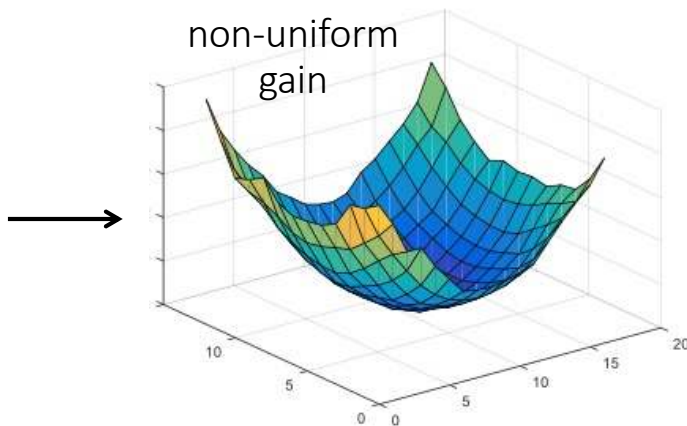
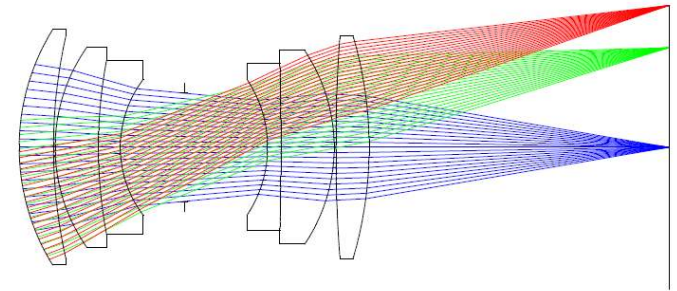


more interesting example of vignetting

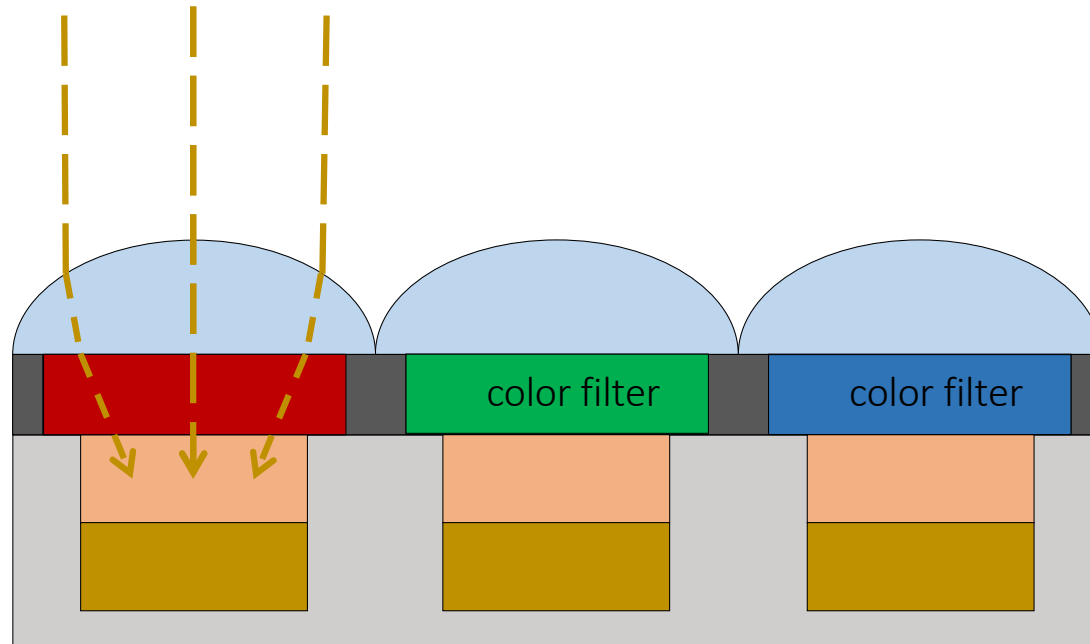
# Vignetting

Four types of vignetting:

- Mechanical: light rays blocked by hoods, filters, and other objects.
- Lens: similar, but light rays blocked by lens elements.
- Natural: due to radiometric laws (“cosine law”).
- Pixel: angle-dependent sensitivity of photodiodes.



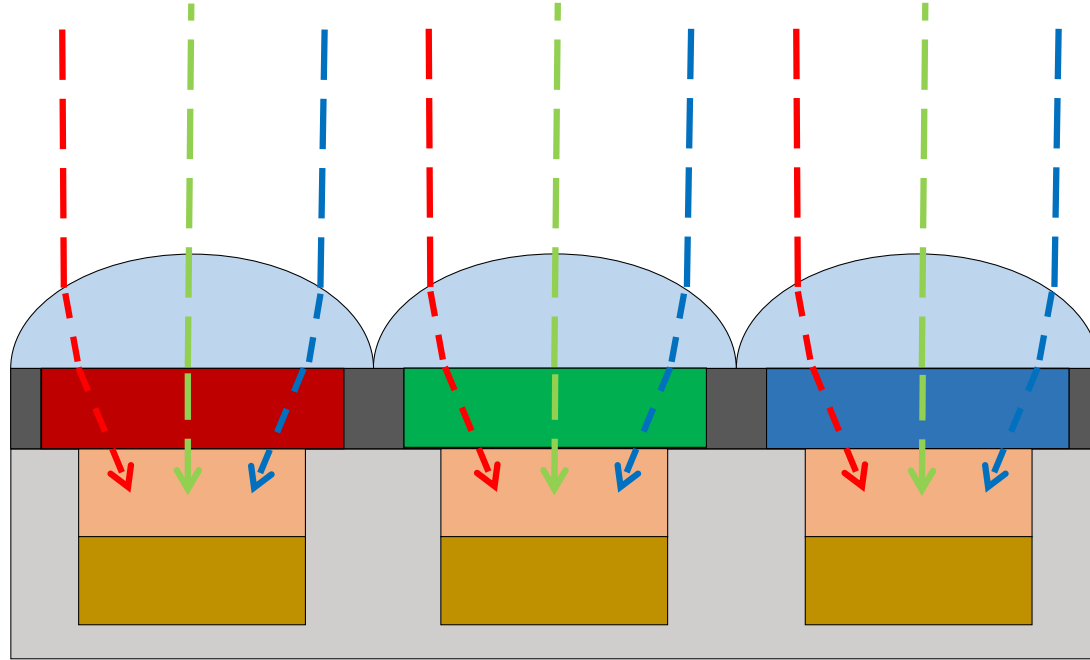
# Color filter



There is no notion of the “color of a photon” only its energy

... but we will color code them for clarity

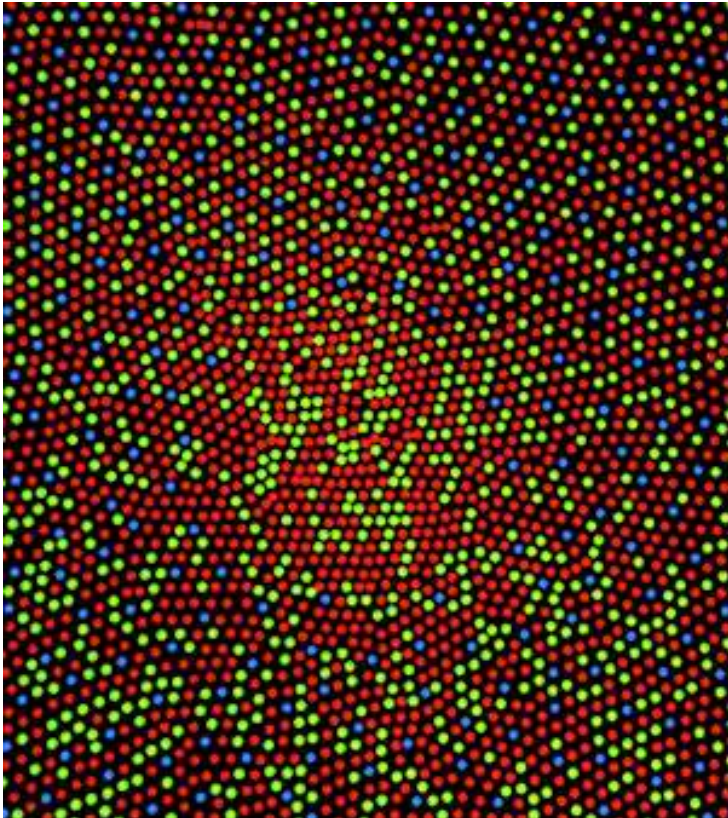
## Color filter



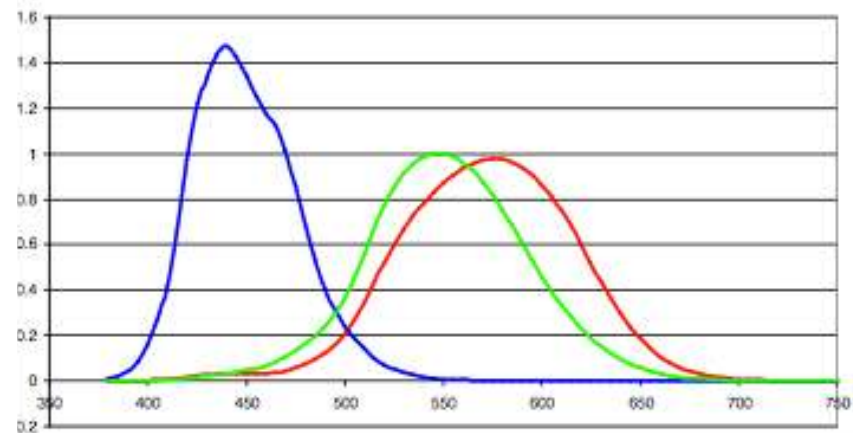
Only lets some light through (according to its spectral density function)



# Spectral Sensitivity Function of Human Eye



Each rod in our eye doesn't record color...  
they are each sensitive to a different  
range of colors



Sensitivity functions of rods in the human eye

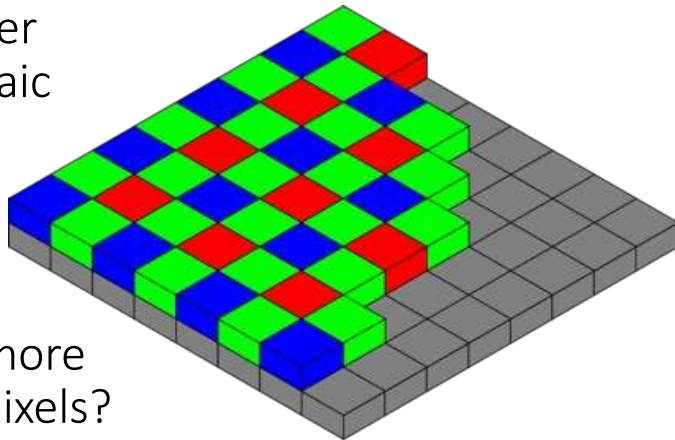
# What color filters to use?

Two design choices:

- What spectral sensitivity functions  $f(\lambda)$  to use for each color filter?
- How to spatially arrange (“mosaic”) different color filters?

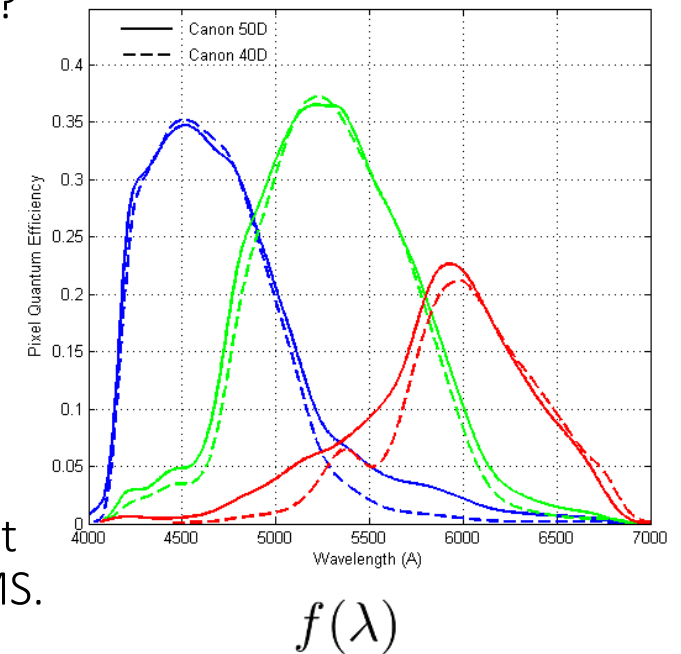
Bayer  
mosaic

Why more  
green pixels?



SSF for  
Canon 50D

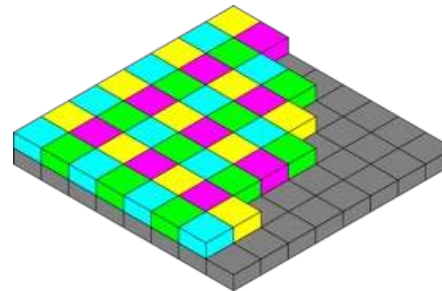
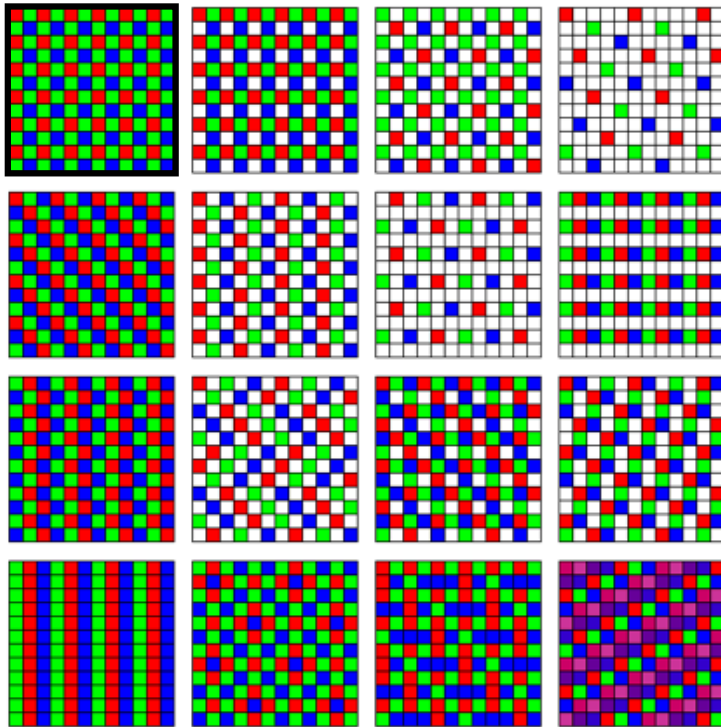
Generally do not  
match human LMS.





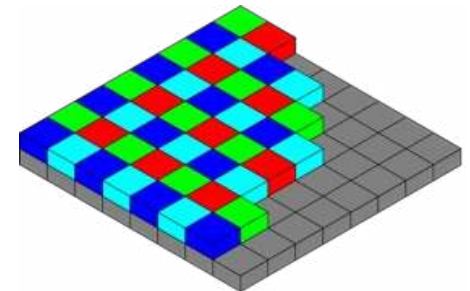
# Many different CFAs

Finding the “best” CFA mosaic is an active research area.



CYGM

Canon IXUS, Powershot

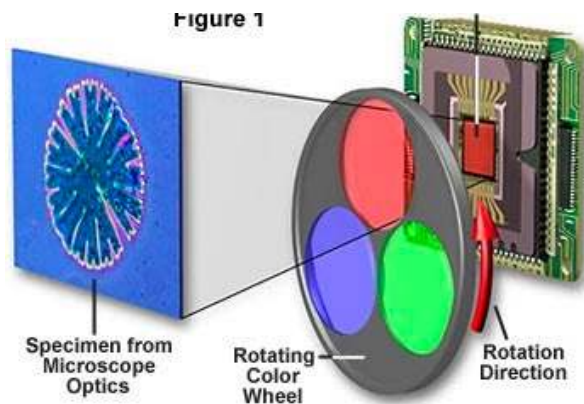


RGBE

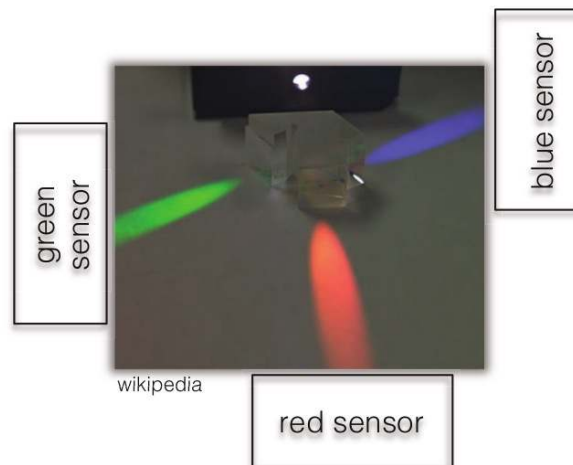
Sony Cyber-shot

# Can you think of other ways to capture color?

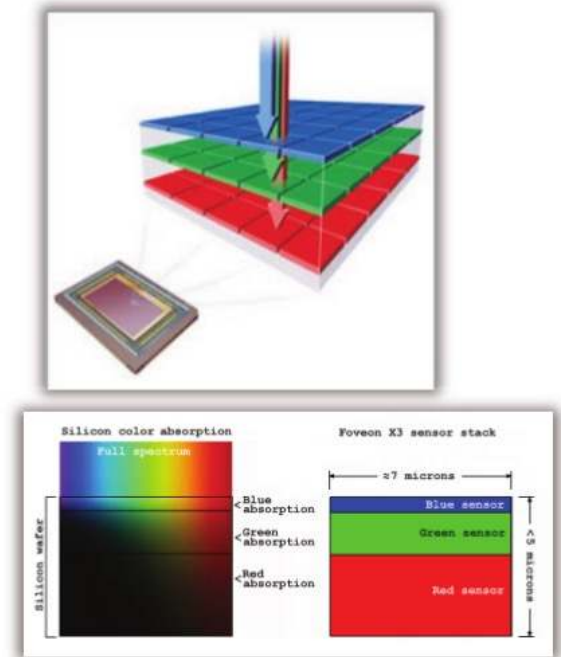
Sequential color filters



Multiple sensors



Vertically stacked

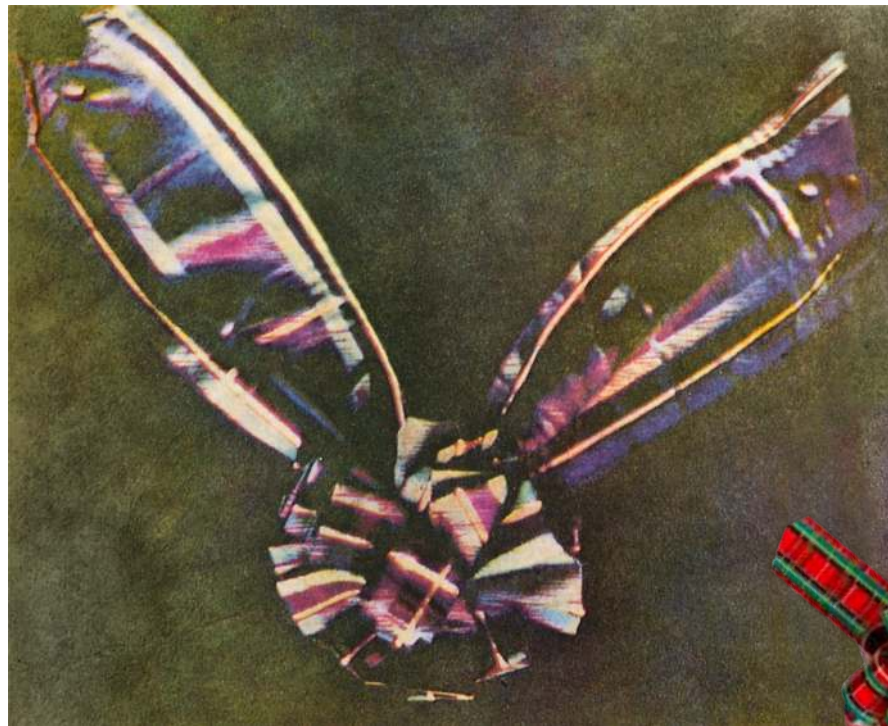


Foveon X3

# Who made the first color photograph?



James Clerk Maxwell

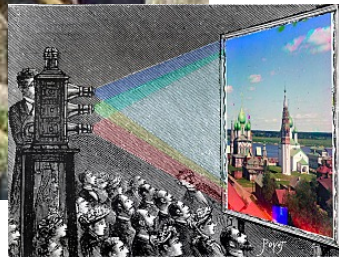
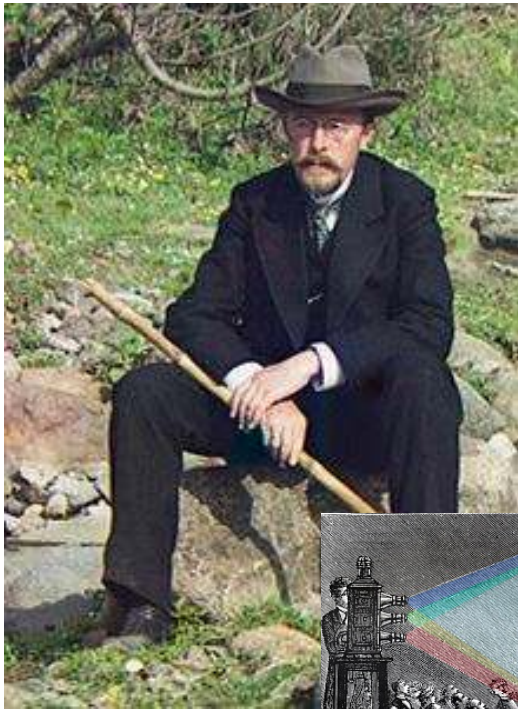


May 17, 1861 – Tartan Ribbon

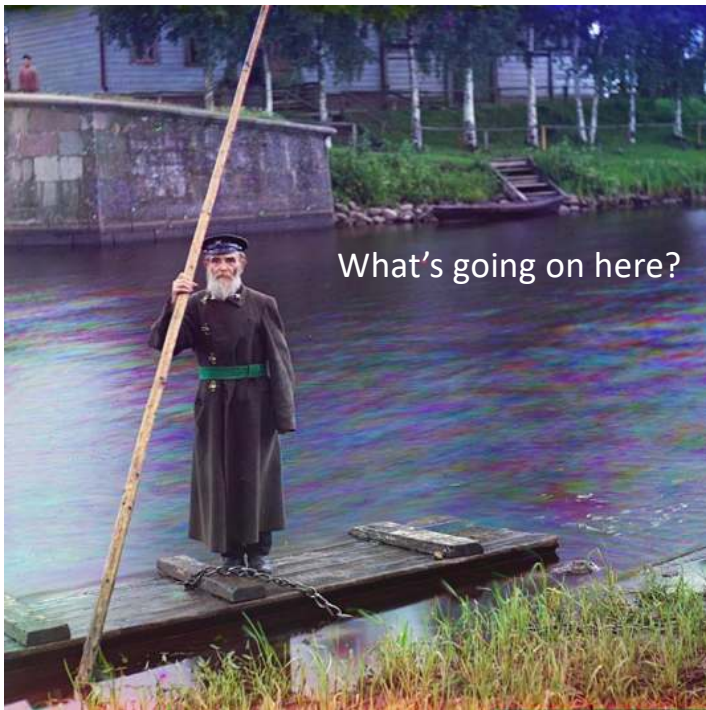




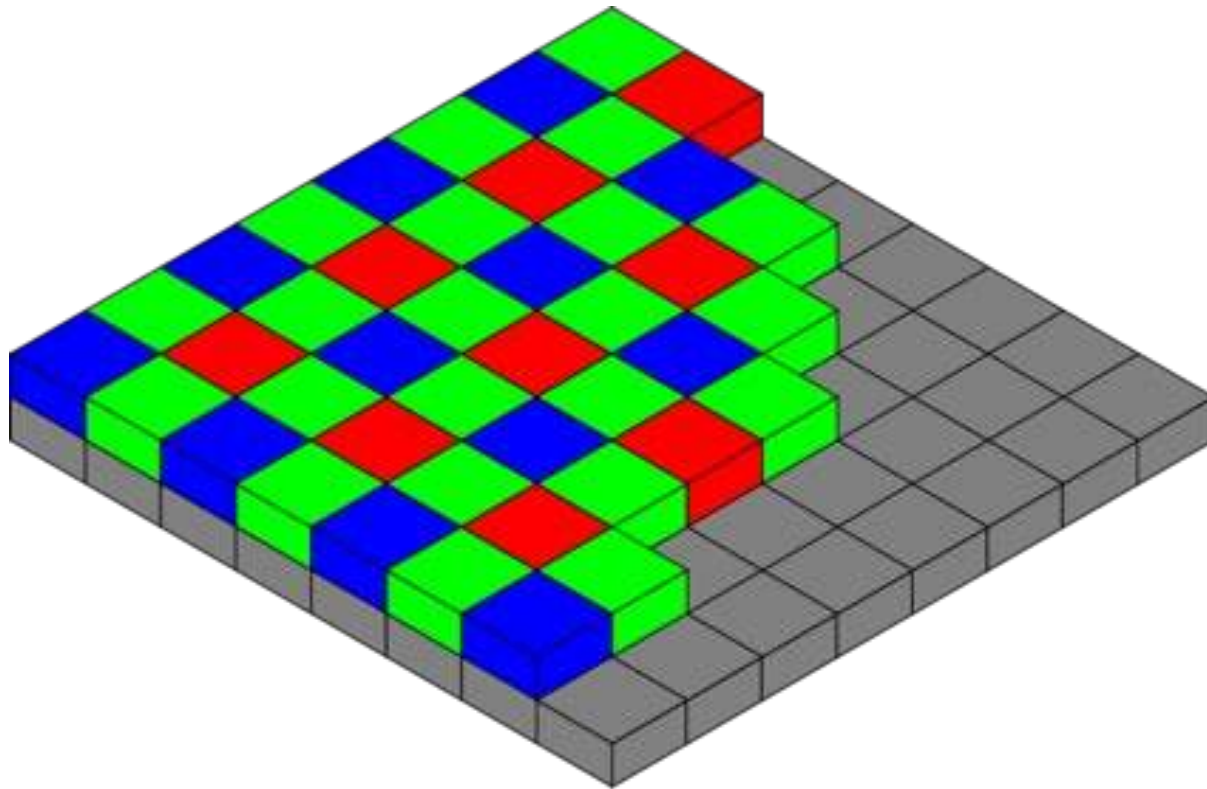
# Sergey Prokudin-Gorsky



# Sergey Prokudin-Gorsky



Most modern cameras use a Bayer filter





After all of this, what does an image look like?



lots of  
noise

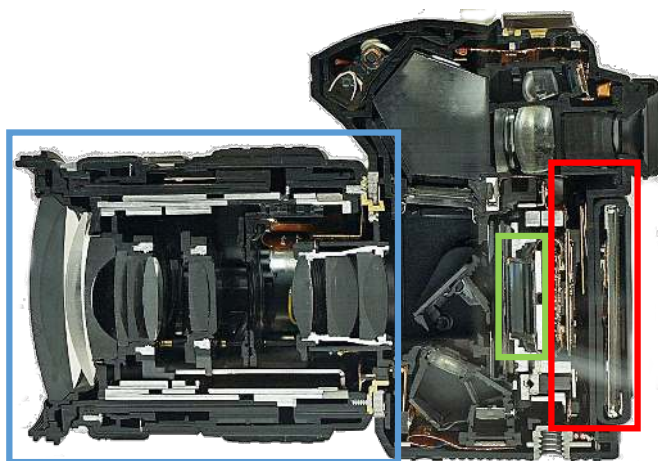


We call this the *RAW* image.

# The modern photography pipeline



post-capture processing



optics and  
optical controls

sensor, analog  
front-end, and  
color filter array

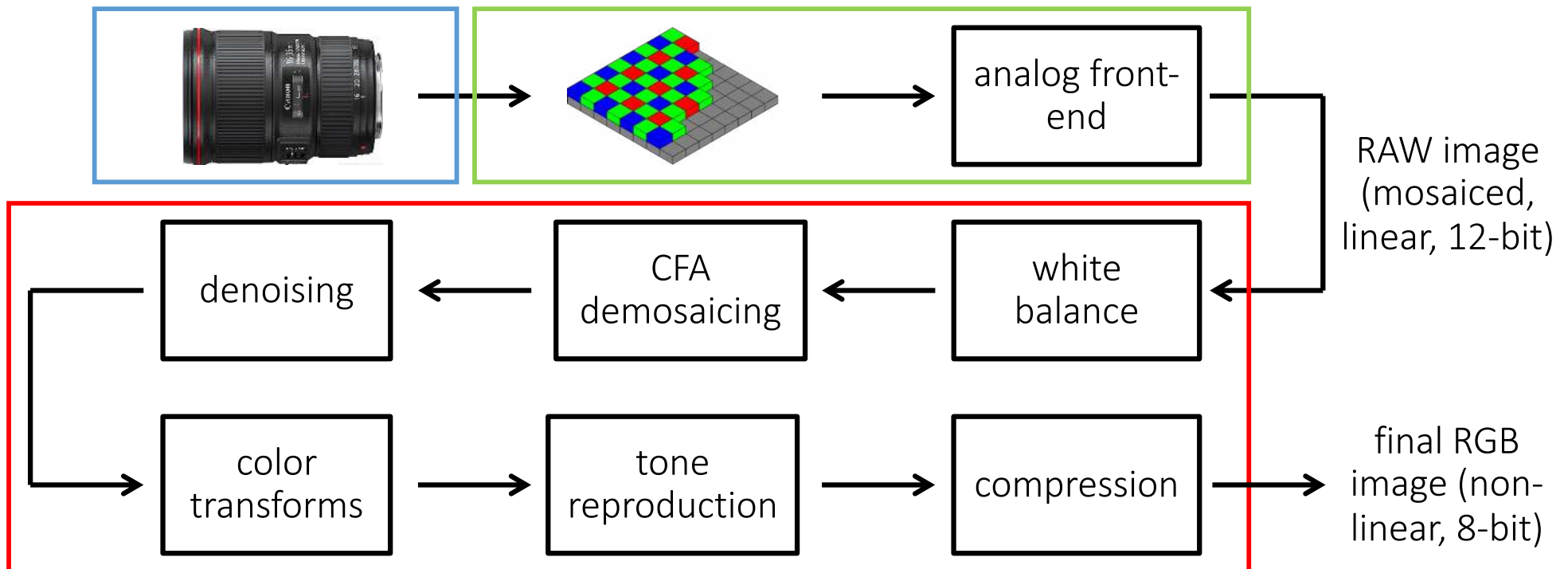
in-camera image  
processing  
pipeline



The in-camera image processing pipeline

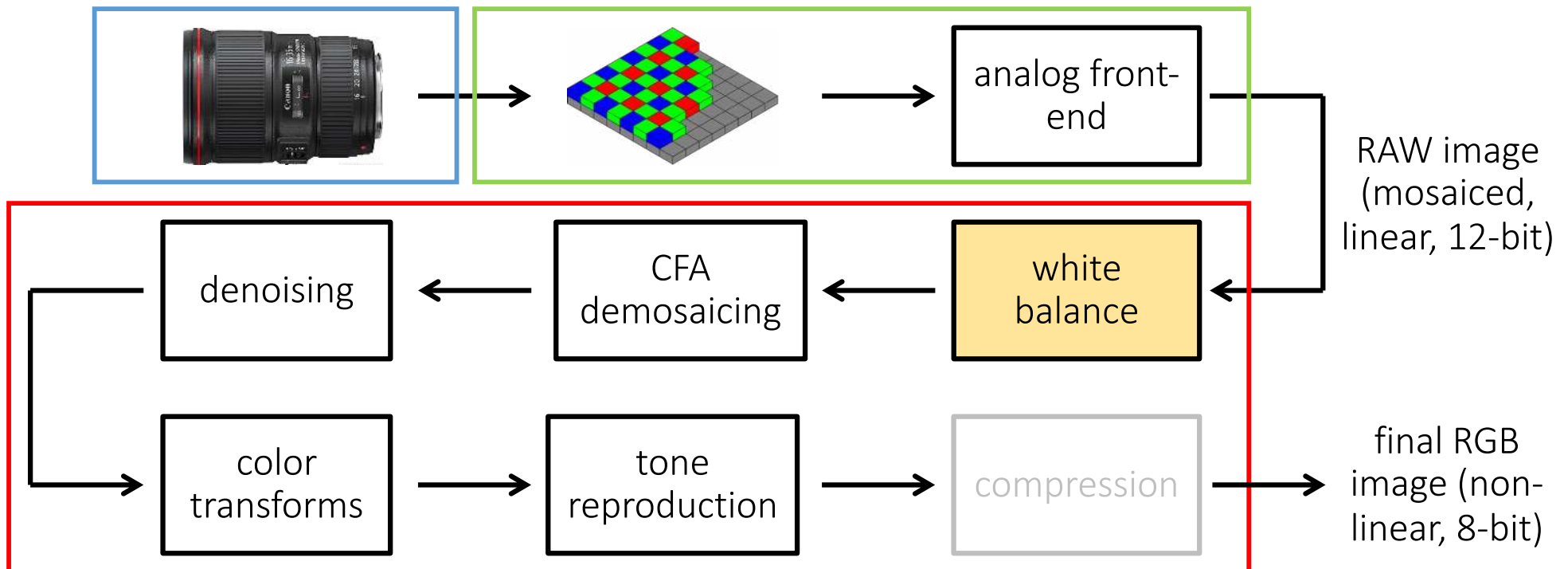
# The (in-camera) image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



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The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



# White balancing

Human visual system has *chromatic adaptation*:

- We can perceive white (and other colors) correctly under different light sources.



# White balancing

Human visual system has *chromatic adaptation*:

- We can perceive white (and other colors) correctly under different light sources.

Retinal vs  
perceived color.



# White balancing

Human visual system has *chromatic adaptation*:

- We can perceive white (and other colors) correctly under different light sources.
- Cameras cannot do that (there is no “camera perception”).

White balancing: The process of removing color casts so that colors that we would *perceive* as white are *rendered* as white in final image.



different whites



image captured  
under fluorescent



image white-  
balanced to daylight

# Manual vs automatic white balancing

Manual white balancing:

- Select a camera preset based on lighting.



Can you think of any other way to do manual white balancing?

# Manual vs automatic white balancing

Manual white balancing:

- Select a camera preset based on lighting.
- Manually select object in photograph that is color-neutral and use it to normalize.



How can we do automatic white balancing?



# Manual vs automatic white balancing

Manual white balancing:

- Select a camera preset based on lighting.
- Manually select object in photograph that is color-neutral and use it to normalize.



Automatic white balancing:

- Grey world assumption: force average color of scene to be grey.
- White world assumption: force brightest object in scene to be white.
- Sophisticated histogram-based algorithms (what most modern cameras do).

# Automatic white balancing example



input image



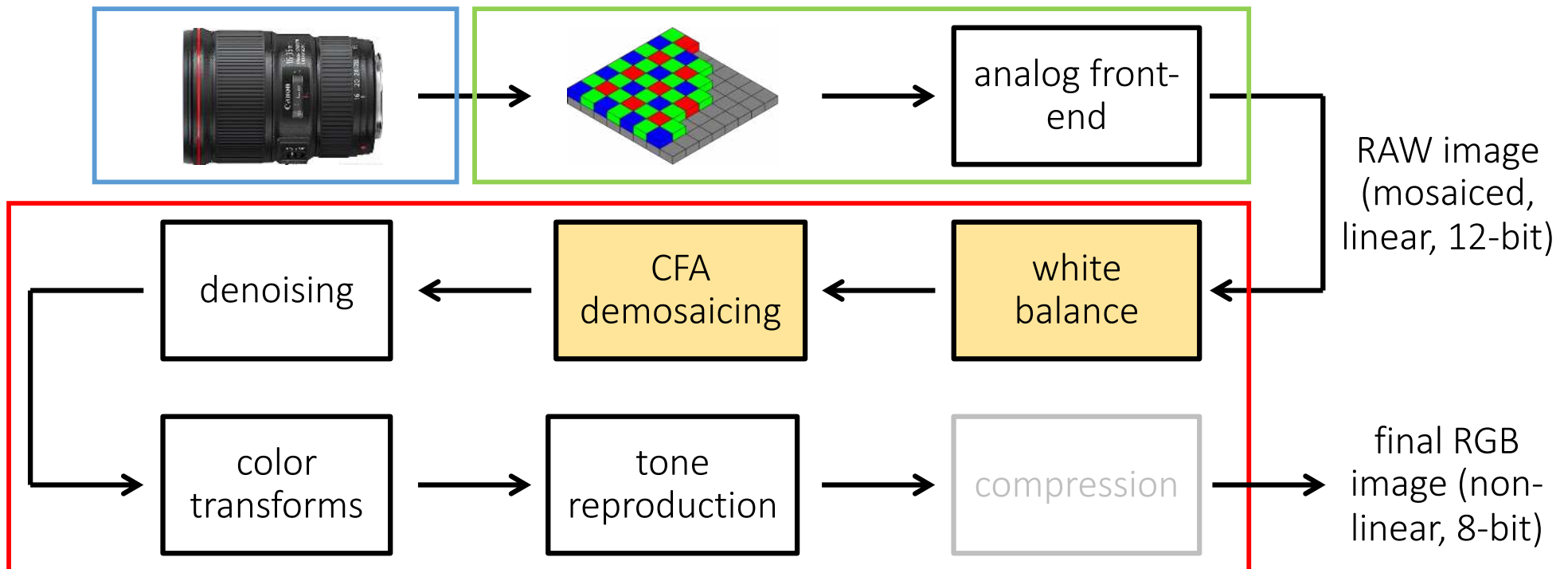
grey world



white world

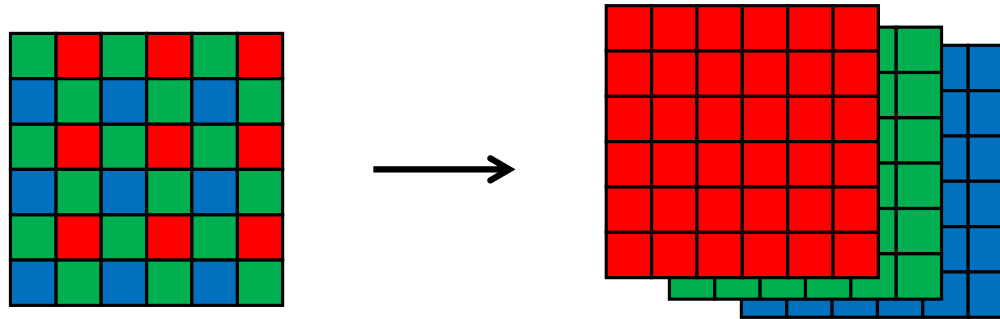
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# CFA demosaicing

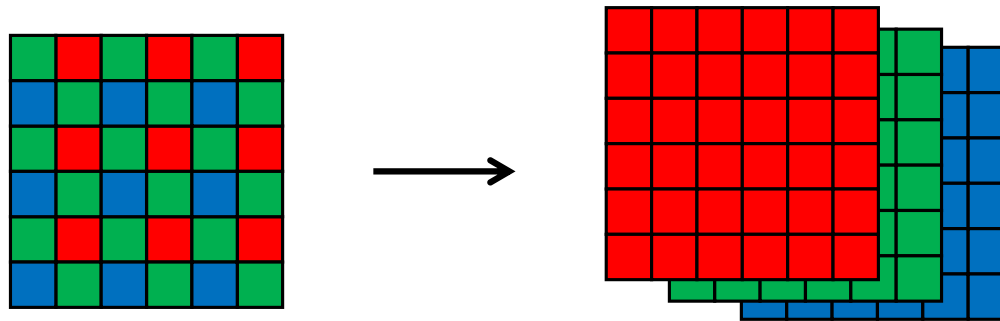
Produce full RGB image from mosaiced sensor output.



Any ideas on how to do this?

# CFA demosaicing

Produce full RGB image from mosaiced sensor output.



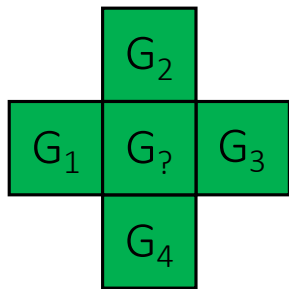
Interpolate from neighbors:

- Bilinear interpolation (needs 4 neighbors)
- Bicubic interpolation (needs more neighbors, may overblur)
- Edge-aware interpolation

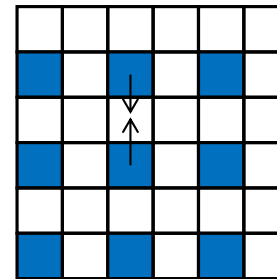
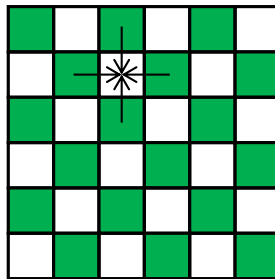
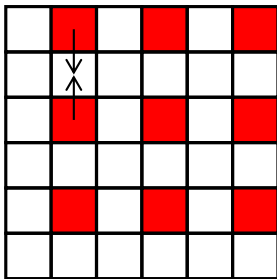
2/3 of the pixel values are made up!

# Demosaicing by bilinear interpolation

Bilinear interpolation: Simply average your 4 neighbors.

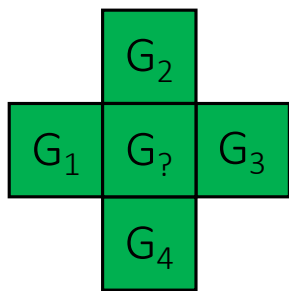

$$G_{?} = \frac{G_1 + G_2 + G_3 + G_4}{4}$$

Neighborhood changes for different channels:

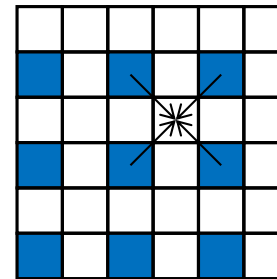
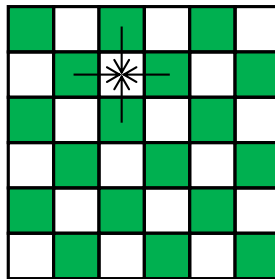
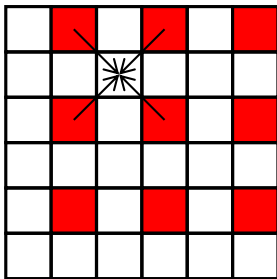


# Demosaicing by bilinear interpolation

Bilinear interpolation: Simply average your 4 neighbors.

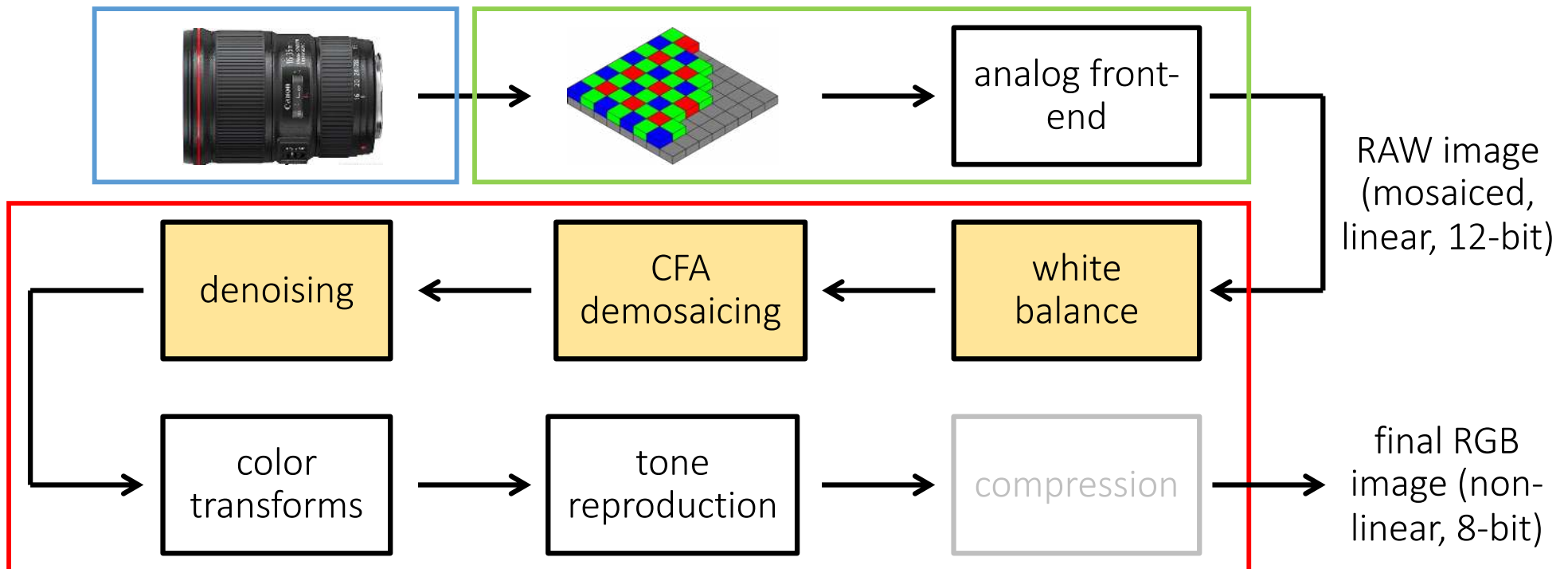

$$G_{?} = \frac{G_1 + G_2 + G_3 + G_4}{4}$$

Neighborhood changes for different channels:



# The (in-camera) image processing pipeline

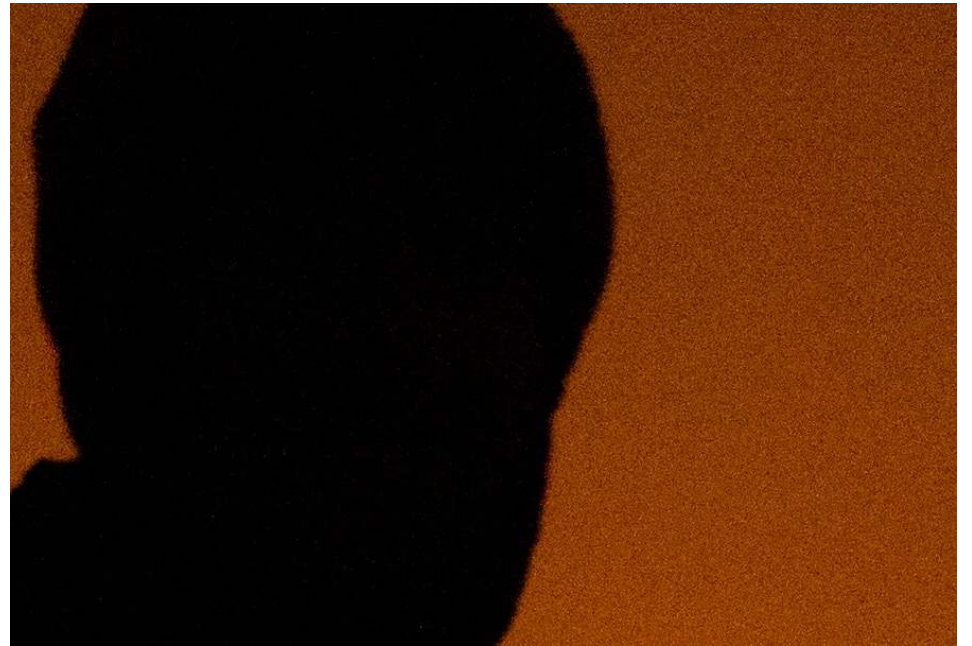
The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



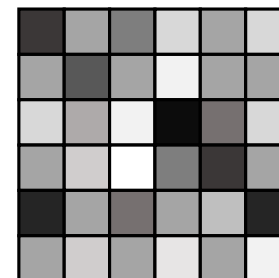


# Noise in images

Can be very pronounced in low-light images.

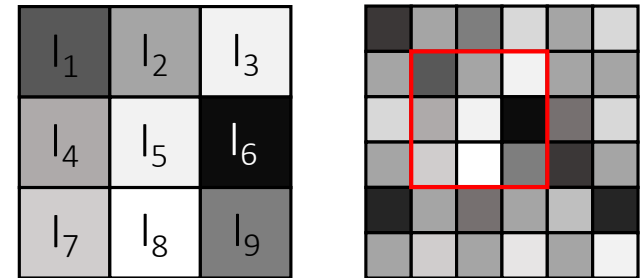


How could you denoise?



# How could you denoise?

Look at the neighborhood around you.



- Mean filtering (take average):

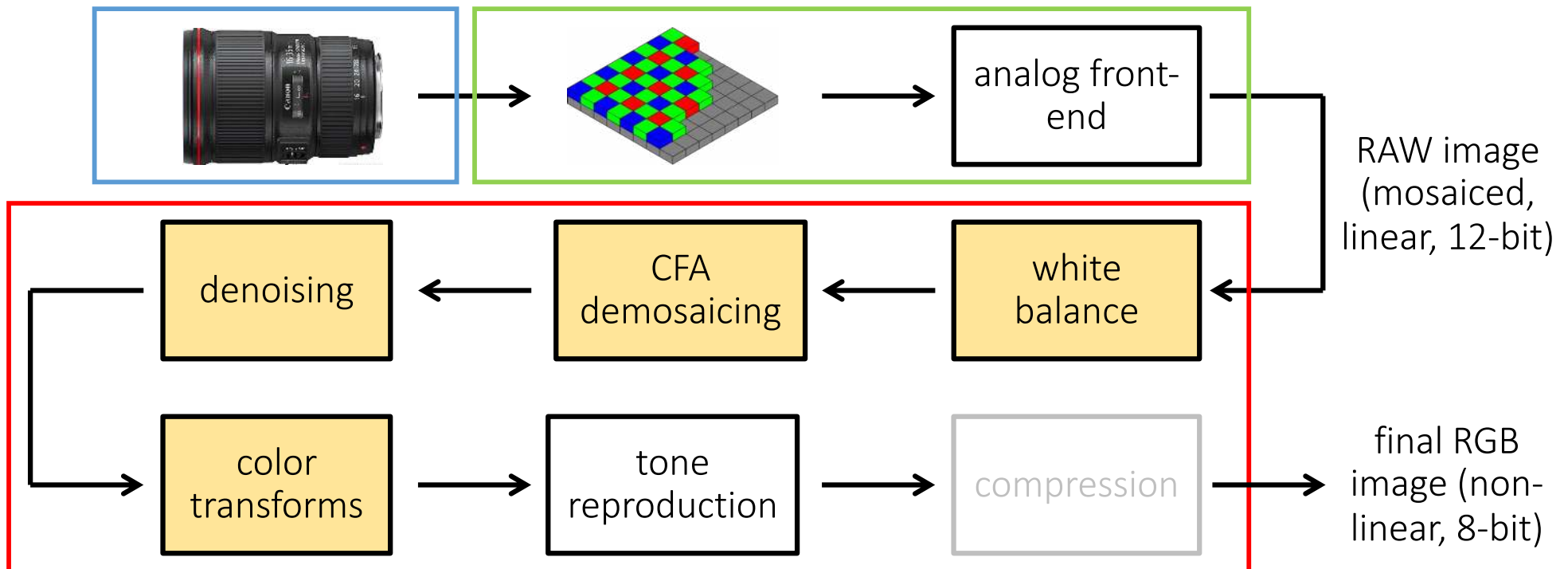
$$l'_5 = \frac{l_1 + l_2 + l_3 + l_4 + l_5 + l_6 + l_7 + l_8 + l_9}{9}$$

- Median filtering (take median):

$$l'_5 = \text{median}(l_1, l_2, l_3, l_4, l_5, l_6, l_7, l_8, l_9)$$

# The (in-camera) image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



# sRGB (standard RGB)

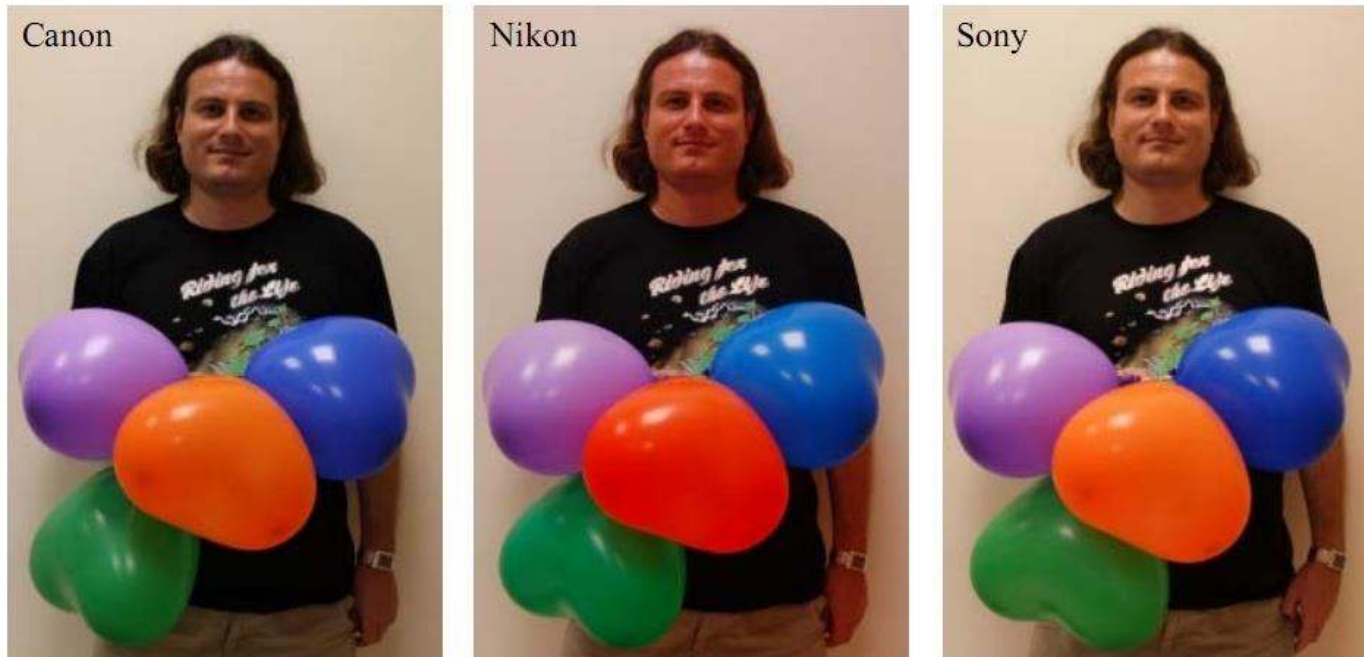


*A (sometimes nonlinear)  
standardized color space  
built for the digital world .... which ends up being not so standard*

# Many different spectral sensitivity functions

Each camera has its more or less unique, and most of the time *secret*, SSF.

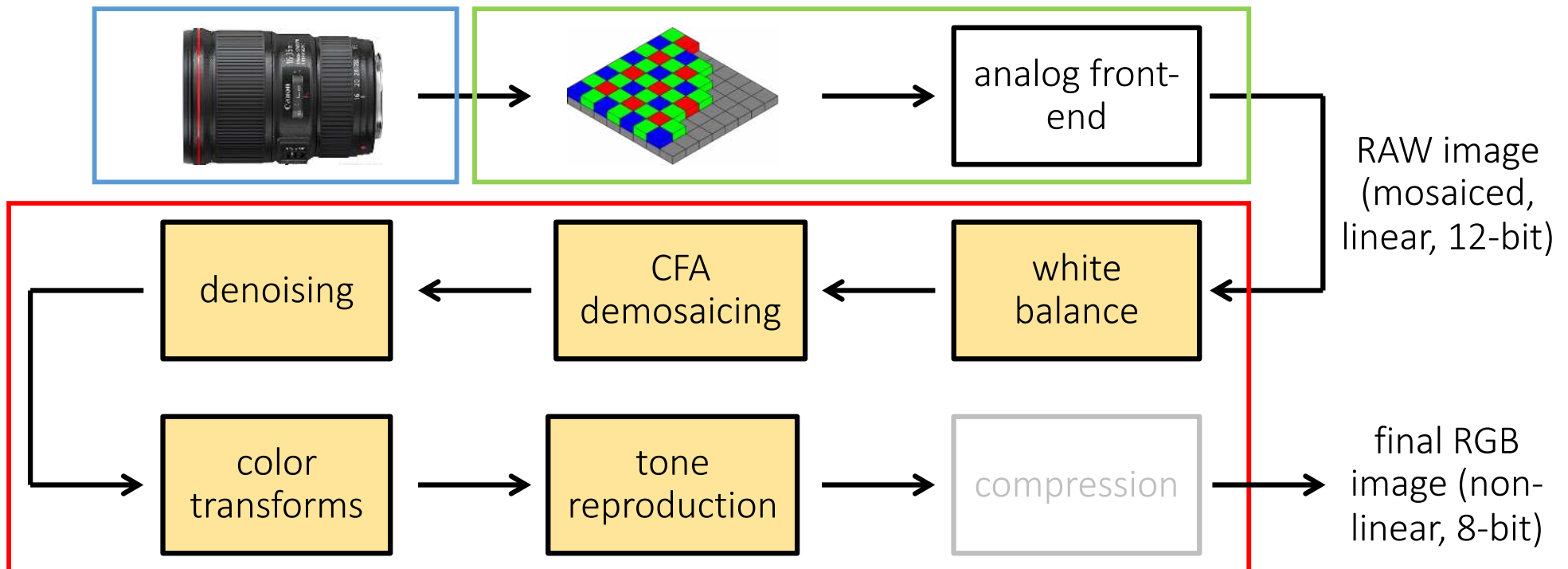
- Makes it very difficult to correctly reproduce the color of sensor measurements.
- We will see more about this in the color lecture.



Images of the same scene captured using 3 different cameras with identical settings.

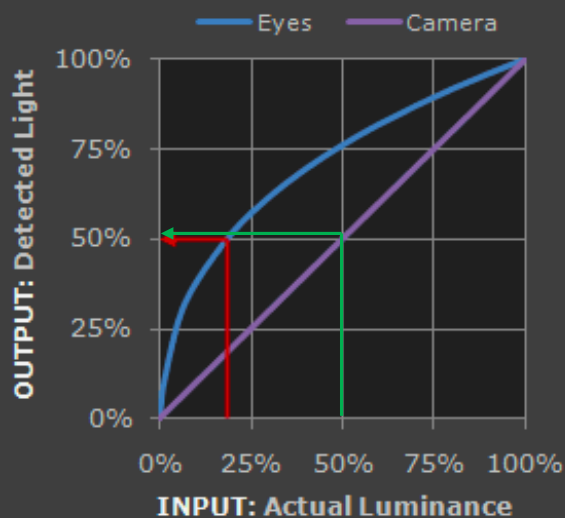
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# Perceived vs measured brightness by human eye

- Sensor response is linear
- Human-eye *response* (measured brightness) is also linear, however *perception* (perceived brightness) is *non-linear*



Reference



Perceived as 50% bright by human

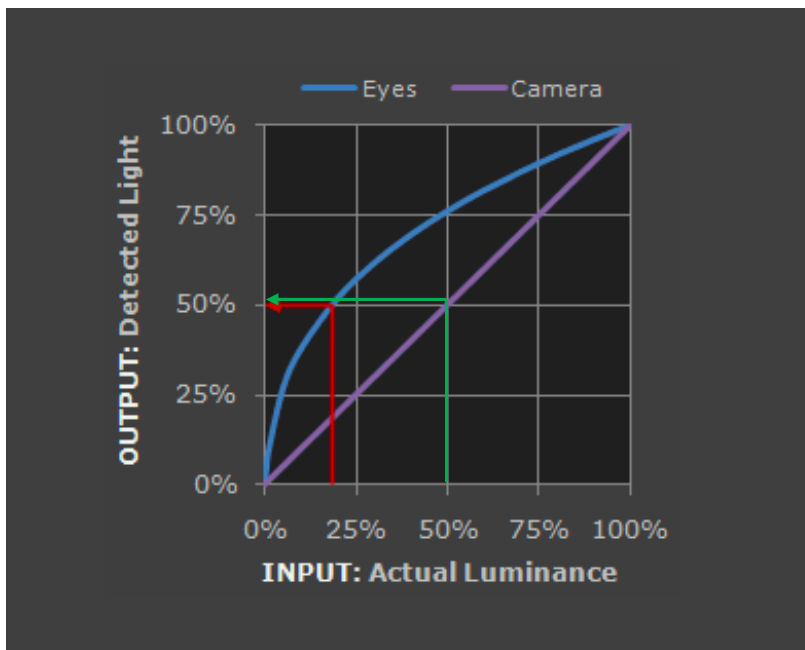


Measured as 50% bright by sensor



# Perceived vs measured brightness by human eye

- Sensor response is linear
- Human-eye *response* (measured brightness) is also linear, however *perception* (perceived brightness) is *non-linear*



More sensitive to changes in lights or darks?



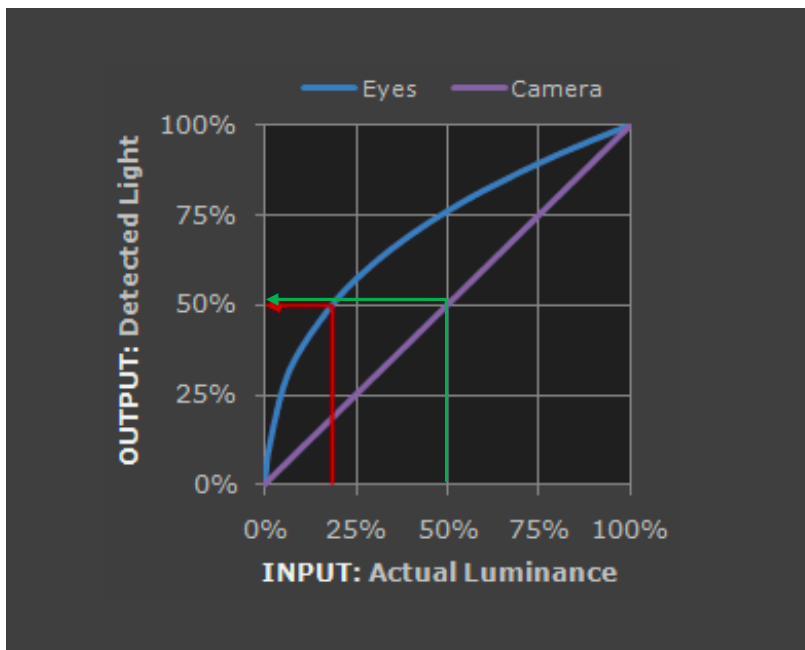
lights



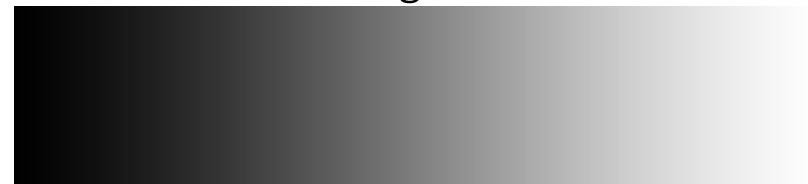
darks

# Perceived vs measured brightness by human eye

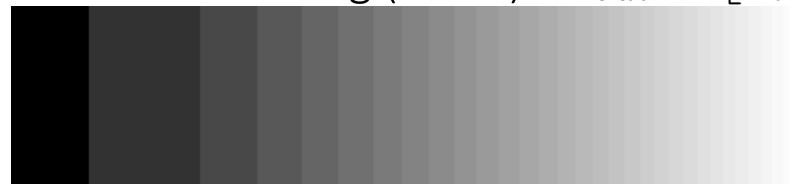
- Sensor response is linear
- Human-eye *response* (measured brightness) is also linear, however *perception* (perceived brightness) is *non-linear*



original



linear encoding (5 bits)  $V_{out} = \lfloor V_{in}/2^5 \rfloor$

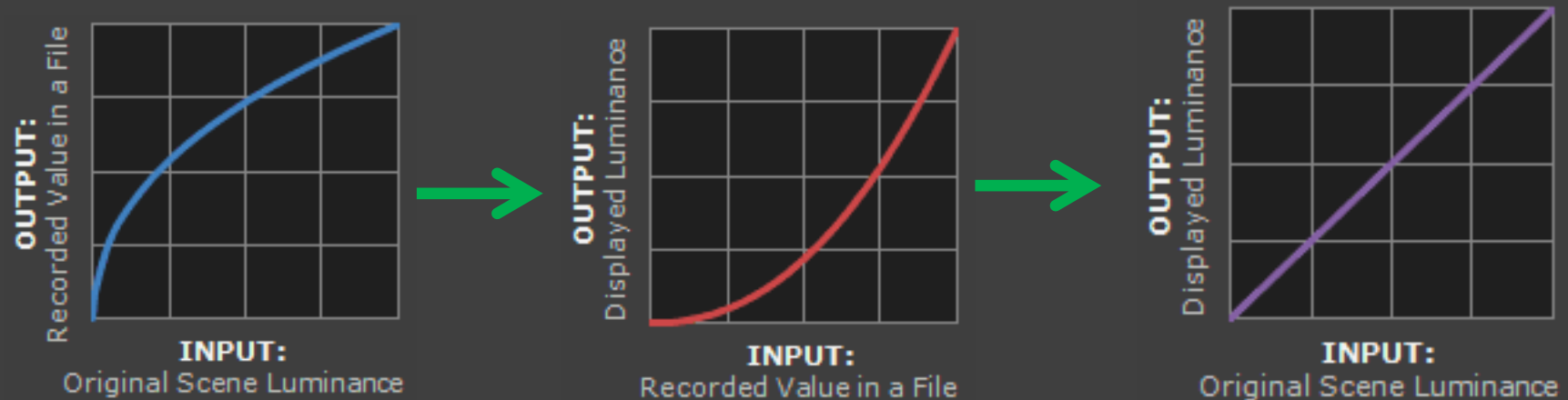


gamma encoding (5 bits)  $V_{out} = \lfloor V_{in}^\gamma/2^5 \rfloor$



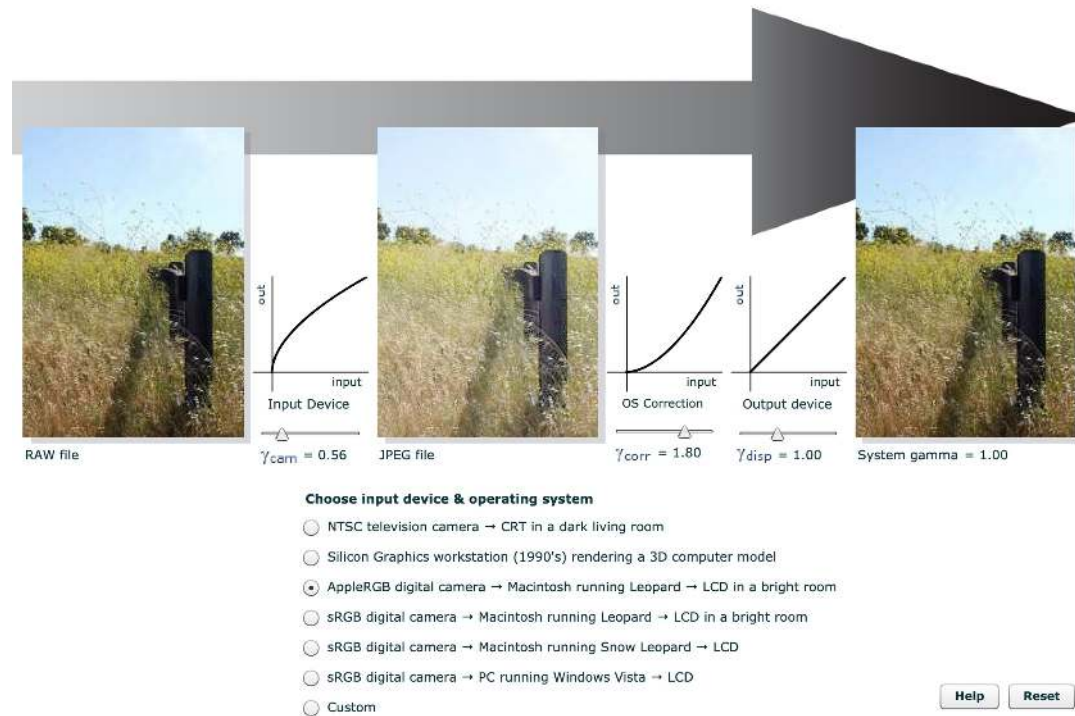
# Gamma encoding for *recording* light

- Gamma encoding happens prior to quantization
- A gamma encoded image must be “corrected” when displayed so that the output luminance is linear again
- Luckily, our monitors do that for us



# Applet

- <https://sites.google.com/site/marclevoylectures/applets/gamma-correction>



# Gamma encoding

$$V_{out} = V_{in}^{\gamma}$$



$$\gamma = 1$$

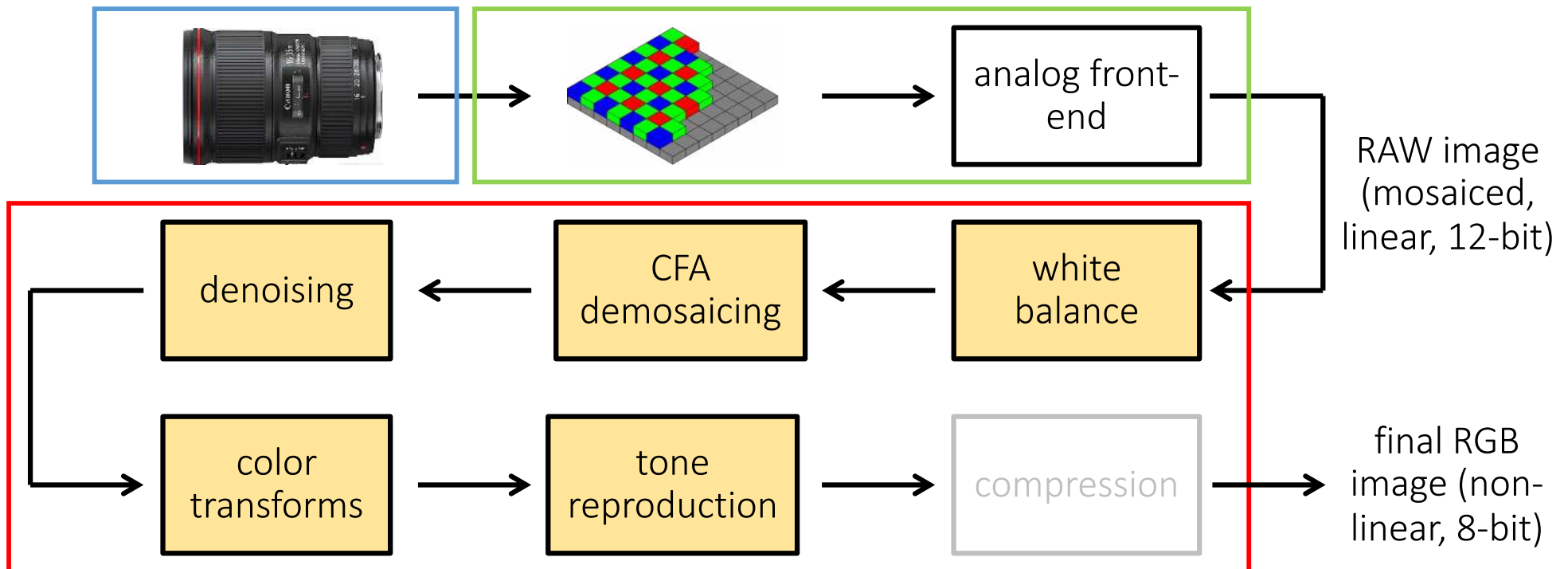


$$\gamma = 1/2.2$$

Warning: looks good, but our values are no longer linear relative to scene radiance!

# The (in-camera) image processing pipeline

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Some general thoughts on the image processing  
pipeline

# Do I ever need to use RAW?

Yes!

- Every time you use a physics-based computer vision algorithm (e.g., almost everything we talk about in this course), you *need linear measurements of radiance*.
- Applying the algorithms on non-linear (i.e., not RAW) images will produce completely invalid results.



# What if I don't care about physics-based vision?

## Still Yes!

You often still *want* to use RAW

- If you like re-finishing your photos (e.g., on Photoshop), RAW makes your life much easier and your edits much more flexible.

# What if I don't care about physics-based vision?

## Still Yes!

You often still *want* to use RAW

- If you like re-finishing your photos (e.g., on Photoshop), RAW makes your life much easier and your edits much more flexible.

## Are there any downsides to using RAW?

Image files are *a lot* bigger.

I forgot to set my camera to RAW, can I still get the RAW file?

Unfortunately, No

- The image processing pipeline is lossy: After all the steps, information about the original image is lost.
- Sometimes we may be able to reverse a camera's image processing pipeline *if we know exactly what it does* (e.g., by using information from other similar RAW images).
- The conversion of PNG/JPG back to RAW is known as “derendering” and is an active research area.

## Take-home message

The values of pixels in a photograph and the values output by your camera's sensor are two very different things.

The relationship between the two is complicated ...and often unknown.

Questions?